

Morning Broadsheet

The visual re-founding of myStockMarket. This document is the design authority for everything the reader sees; DEVELOPMENT-PLAN.md remains the authority for everything the product computes. Where the two disagree on looks, this plan wins. Where anything disagrees on honesty, the honesty rule wins — always.

“Every amended rule is about how the product feels; every preserved rule is about whether the product can lie. Shadows and gradients cannot make a claim; a projected line, a bare percentage, or an animated probability can.”

— Part 2, the line between taste and honesty

How this document binds

Executor: Claude Opus 4.8, unattended, same working rules as DEVELOPMENT-PLAN.md (TDD per §6.2 there, plain-English code and docs, session ritual, DECISIONS/LESSONS logging). **Source design:** `FigmaDesignRef/Stock Market Command Center/` — a Figma Make export (React + Tailwind v4) whose `src/index.css` is the token source and whose twelve components are the screen references. The screenshot in `src/imports/` is mood, not spec. **Authority order for this work:** honesty rules (unchanged, listed in Part 2) > this plan > the Figma export > DEVELOPMENT-PLAN.md Part 3 (now amended) > judgment.

Decisions I need from you

Four choices ripple through everything. Each has a recommendation and the plan is written assuming the recommendation, so answering "all as recommended" costs you one sentence. Every other decision in this redesign is local; I made it and logged it in Appendix E.

D1 — Dark Desk: keep it, re-skinned. (Recommended: KEEP) The Figma is light-only, but the Dark Desk shipped in P6 and the toggle exists. I recommend keeping dark mode Desk-only and re-deriving it from the new palette ("Midnight Desk" — deep indigo-night glass, same geometry, dimmed orbs; full token column in Appendix A). Cost: every color token needs a dark twin and the visual-regression matrix doubles for Desk pages. The alternative — dropping dark mode — deletes a working P6 feature and its e2e coverage. **Ripples:** token sheet, VRT matrix, `lib/tokens.ts` status-bar colors, theme e2e.

D2 — Mobile navigation: bottom tab bar on phones. (Recommended: BOTTOM TAB BAR.) You asked me to keep the top navigation and make its mobile collapse genuinely better. On desktop the top bar stays (restyled, same restraint). On phones I recommend moving the five rooms into a fixed glass bottom tab bar (Desk · Scans · Paper · Track · Academy — icons + labels, 44px+ targets, safe-area padded), with Settings behind a gear in the top bar. This is the standard PWA ergonomic answer: rooms live where thumbs are, nothing hides off-screen, and the top bar shrinks to brand + date + gear. The alternative (keep the top scroll-strip, add edge-fade + auto-centering of the active room) is specced in §4.2 as Plan B — it stays one-handed-hostile for the far rooms, which is why it is Plan B. **Ripples:** shell layout, e2e nav spec, safe-area work, VRT phone baselines.

D3 — Wordmark: keep the name, adopt the mark. (Recommended: KEEP "myStockMarket" + new mark.) The Figma brands itself "Market Desk" with a 28px indigo-gradient "M" logomark. Renaming the product is your call, not mine. I recommend keeping the name and adopting the mark: gradient "M" tile + "myStockMarket" in tracked mono caps. If you prefer "Market Desk" (it reads better in the split-screen login), say so — it touches the nav, login, manifest, PWA icons, and copy. **Ripples:** nav, login, `manifest.ts`, PWA icon set, README/docs.

D4 — The two rooms: keep warm/cool contrast, translated. (Recommended: KEEP TWO ROOMS.) The Figma paints one lavender world. Our constitution deliberately makes the Academy a different, warmer room so the switch from "reading data" to "learning" is felt. I recommend keeping that: Desk = cool lavender morning light (the Figma), Academy = warm cream light with rose-gold orbs, same glass grammar, same indigo interactive accents (links/focus never change meaning between rooms). Unifying on lavender everywhere is simpler but erases a product idea the Academy was built around (and the Dark-Desk-only rule leans on it). **Ripples:** Academy token column, academy layout, theme e2e, the design one-liner in CLAUDE.md.

How to answer: a single message like "D1 keep, D2 bottom bar, D3 keep name, D4 two rooms" (or just "all as recommended") unblocks the build. Vetoes change bounded, named sections: D1→Appendix A dark column + §8 R1; D2→§4.2; D3→§4.2/§5.7; D4→§5.6 + Appendix A academy column.

The design vision

What the Figma teaches, what survives from Broadsheet Terminal, and where I overruled the Figma as designer. Read this part first; every later spec is this part made mechanical.

1.1 What the Figma actually is

The export is not a generic dashboard theme. Read closely, it is **our product redrawn in a warmer material** — it keeps mono uppercase mastheads ("MARKET PULSE — PRE-MARKET"), hairline dividers inside cards, tabular numerals, dot arrays with visible misses, base rates written as "61 of 108 — 56.5% (47–65% CI)", a "Typical Range (Not a Target)" panel, a "What Would Weaken This" list, and a disclaimer that is almost word-for-word our scope line. The honesty furniture survived the restyle. What changed is the **material and the light**:

- **Light.** A fixed lavender gradient wash (`#f9f8ff → #eef2ff → #f0f4ff → #faf8ff`, 145°) with two soft radial "orbs" glowing from the corners. The page feels like morning light on paper instead of flat bone.
- **Material.** Cards are translucent white glass (`rgba(255,255,255,0.72)`) + 16px blur, 16px radius, violet-tinted hairline border). Raised surfaces get more opacity, more blur, and a soft violet shadow. Depth exists, but it is atmospheric, not skeuomorphic.
- **Type voice.** Three families: Playfair Display (an editorial serif for titles — the "broadsheet" made literal), Inter for UI, JetBrains Mono for every numeral. Serif titles over mono data is the signature pairing.
- **Color voice.** One indigo accent family (`#6366f1` / `#4f46e5`, plus a `#4f46e5→#7c3aed` brand gradient used exactly twice: logomark and primary button). Gain/loss is **blue/orange** (`#2563eb` / `#ea580c`) — already colorblind-safe. Tier chips are quiet soft-wash pills.
- **Restraint that survived.** No neon, no pure black, color still scarce and semantic, one serif hero per page, whitespace still the main material.

1.2 What survives from Broadsheet Terminal (unchanged laws)

1. **Mono numerals everywhere data appears** — JetBrains Mono replaces IBM Plex Mono, the law does not change. Tabular figures, true minus U+2212, right-aligned numeric columns.
2. **Mastheads** — numbered, uppercase, tracked, with a rule and a right-aligned timestamp. The rule becomes a tinted hairline instead of 2px ink, the anatomy is identical.
3. **One hero figure per view** — the S&P level stays the Desk's only 64px numeral.
4. **Numbers render only through** `components/BaseRate` and `Lib/format` — restyled, never re-implemented. `copy.ts` remains the only source of user-facing strings.
5. **The honesty furniture** — N + CI, natural frequency, always-up baseline, decay stamps, weakeners, misses-first, catalyst-or-noise, timestamps everywhere.
6. **Token discipline** — one token sheet in `globals.css`, no ad-hoc hex anywhere else, `/styleguide` as the living spec, anti-drift greps at every phase exit (grep list v2, §3.10).
7. **Calm foundations** — no urgency theater, no autoplaying motion loops, no toasts, no badges, `prefers-reduced-motion` zeroes everything.

1.3 Where I overruled the Figma (designer's calls, not tracing)

#	Figma shows	This plan does instead	Why
1	Mini dot array scaled to 40 dots representing N=108 ("68" tail)	Cards get a proportion bar + printed counts ; the full every-dot-is-a-case array lives in the expanded detail	A scaled dot array misrepresents sample size — the whole point of dots is that you can count them. Honesty rule beats Figma.
2	<code>background-attachment: fixed</code> on the body gradient	A fixed-position wash layer (<code>.app-wash</code>) behind content	<code>background-attachment: fixed</code> is broken on iOS Safari — the PWA would repaint the gradient on every scroll frame.
3	Glass blur on every card	Blur budget: backdrop-filter only on sticky bars + overlays; cards are translucent without blur on phones	Stacked <code>backdrop-filter</code> is a GPU tax that shows up as scroll jank on mid-range phones. Over a static wash, translucency without blur is visually near-identical.
4	Pulsing emerald status dot with glow in the nav	Static dot + mono text label	A glowing live-status dot is urgency theater; session state is peripheral information.
5	Recharts for sparklines/scatter	Keep hand-rolled SVG + lightweight-charts	No new chart dependency; our SVG components already exist and are unit-tested. Restyle, don't replace.
6	AM Plan / PM Scorecard tabs on the brief	Our brief keeps its labeled slots; the evening scorecard stays its own module	The tabs assume a two-briefing product; ours is one evening pipeline (9:00pm ET promise). Don't redesign the product to fit a mock.
7	"Next 21 days" calendar header	"Next 14 days" (the ingest window)	Never claim a window the pipeline doesn't fetch.
8	Emerald/amber tier text at ~3.7:1 contrast	Same hues, darkened one step (AA table in §3.3)	Tier words must clear 4.5:1 — the honesty layer is never the least legible thing.
9	Options-expiry (OPEX) rows in the calendar	Omitted	Not in our product's catalyst model; adding data sources is out of scope for a redesign.
10	Track-record hit/miss in blue/orange text only	Blue/orange chips with the word inside	Outcome must never rely on color alone (colorblind rule) — the word is the redundant channel.
11	Login backed by <code>useState(false)</code>	Our real login wall (<code>proxy.ts</code>) unchanged	Auth is not a styling concern; only <code>LoginForm</code> and the page shell restyle.

#	Figma shows	This plan does instead	Why
12	Figma's <code>--color-warn #d97706</code> free-floating; tier chips in green/amber/gray	The entire amber–orange band is reserved for down/loss and the two alert consumers; tier-moderate and grade-mixed move to teal (§3.3)	Reservation is perceptual, not hex-deep: a Desk full of amber "moderate" chips would drown the gate flag even if no chip shared its exact value.
13	Accent-indigo on small sub-labels, muted gray on the big mastheads/eyebrows (see <code>MarketPulse.tsx:23</code>)	Same: module mastheads stay muted; accent appears only on interactive text and the small in-card slot labels ("WHAT FIRED", brief slots)	Eight indigo mastheads would make the accent mean "chrome" as often as "you can act here". The Figma got this right; an earlier draft of this plan inverted it and was corrected in review.
14	Orbs frozen at the mock's coordinates (480px top-right, 380px bottom-left, composed for a short single-column frame)	Orbs re-derived for a long-scroll two-column product: viewport-ambient light, resized/repositioned per breakpoint, opacity halved on phone (§3.4, Appendix A)	The wash and orbs survive translation because they are <i>light</i> , not content — viewport-anchored ambience is what morning light is. Their geometry did not survive; it was composed for a different frame.
15	Progressive left-to-right chart animation (Recharts default)	Charts appear as completed wholes (200ms opacity fade at most); no directional draw-in anywhere	A left-to-right sweep on a price series reads as momentum — "...and then" — priming a directional expectation right next to visuals that insist they are not forecasts.

1.4 The one-liner (replaces "Broadsheet Terminal" in CLAUDE.md)

"Morning Broadsheet": editorial serif over mono numerals, morning-light lavender wash, glass cards with soft depth, hairlines inside cards, one hero figure, honesty furniture on every claim, two rooms (cool lavender Desk / warm cream Academy). Color is scarce and always means something. If it could be a template — austere or glossy — it is wrong.

The constitution: what is amended, what is preserved, and why

The user is deliberately relaxing the aesthetic rules and deliberately keeping the honesty rules. This part is the legal text: every amended clause and every preserved clause, with the file and line it lives in, so no document contradicts another. The doc edits themselves are executed with this plan (Part 10) — Opus never has to reconcile two constitutions.

2.1 AMENDED — aesthetic rules relaxed by user directive (2026-07-11)

#	Old rule (verbatim anchor)	Lives in	New rule
A1	"no card shadows, no decorative borders" (RR §9.7); "drop shadows on cards (the drawer scrim is the sole shadow)" (plan §3.1 banned list)	<code>docs/src/rr-04.html</code> §9.7, <code>DEVELOPMENT-PLAN.md</code> §3.1/§3.4	Soft elevation is allowed through shadow tokens only (<code>--shadow-soft/--lift/--overlay</code> , §3.4). Arbitrary shadows stay banned by grep.
A2	"gradients and glassmorphism" banned (plan §3.1)	same	Gradients and glass allowed through gradient/surface tokens only (<code>--gradient-brand/--wash/--range</code> , glass surfaces §3.4). Raw <code>linear-gradient()</code> outside <code>globals.css</code> stays banned by grep.
A3	" <code>rounded-2xl</code> softness — the app's radius is 2px" (plan §3.1, §3.4)	same	Radius scale: 16px cards, 12px panels, 10px controls, 8px chips, pill for tabs/dots (§3.5). Arbitrary <code>rounded-[...]</code> stays banned.
A4	"colored-chip confetti" banned; tiers render as neutral gray chips (plan §3.1, RR §9.7)	same	Colored soft-wash chips allowed with AA text and a non-color redundant channel (the word itself). Tier chips get color (§3.3). "Confetti" (color without meaning) stays banned in the checklist.
A5	"purple anything" banned (plan §3.1)	same	Indigo/violet is now the accent family. The ban is repealed outright.
A6	Motion: 160/200ms fades only; "nothing else moves" (plan §3.5)	same	General UI motion allowed per the motion system (§3.6): page transitions, hover lifts, drawer slides, chart draw-in, micro-interactions. The probability/money stillness rule survives verbatim (P6 below).
A7	"Data refreshes fade in over ~500ms" / calm-tech framing of §9.7	<code>rr-04.html</code> §9.7	Calm is preserved as no urgency, no loops, no autoplay — not as "no motion." Rewritten in the amendment note.
A8	Skeleton shimmer banned (plan §3.1)	same	A single quiet shimmer (1.6s, low-contrast, reduced-motion-aware) is allowed on loading placeholders. Never on probability/money placeholders — those load as text.
A9	"If it could be a default template, it is wrong" aimed at SaaS gloss	CLAUDE.md one-liner; plan §3.1	Kept, but re-aimed at both failure modes: default-austere and default-glossy (§1.4).
A10	Dark values #131412 family ("dark bone")	<code>globals.css</code> , plan §3.3 dark column	Re-derived as "Midnight Desk" (deep indigo night), still Desk-only (D1/D4).

2.2 PRESERVED — honesty rules that survive unchanged (non-negotiable)

#	Rule	Lives in (unchanged)	Redesign consequence
P1	No point predictions. No single projected line for where a price is headed. Forecasts are range bands with frequency labels and quantile dotplots.	RR §8.1, §9.8; plan §15.1	The range-band ladder becomes the app's <i>visual centerpiece</i> (§5.5) — beautiful, layered, still honest. It has no 50th-percentile mark of any kind (no stroke, no tick, no label at the median); bands are labeled by coverage frequency only. And it is deliberately NOT drawn as a forward-widening cone (§3.8) — a cone silhouette reads as a projection even when labeled historical.
P2	No motion on probability visuals. Base rates, dot arrays, CI bars, calibration charts, vol/range bands, Brier figures render complete on first paint. No draw-in, no easing, no ticking numbers — ever. No ancestor may move them either: no transform, reveal animation, or transition on any element containing a P2 component. The single sanctioned ancestor animation is the uniform route fade (§3.6) — opacity-only, no translation, every frame shows the visual complete and unmoving relative to everything else.	plan §3.5 (kept clause), §9.7 "probability visuals are motionless"	Every P2 component root carries a <code>data-p2</code> attribute; the motion system (§3.6) carves them out, the grep v2 checks their files, and a DOM-level test walks up from every <code>[data-p2]</code> node asserting no animated/transformed ancestor (§3.10). General UI motion never leaks in — not even from wrappers.
P3	No composite buy/sell gauges or "Technical Rating: BUY" aggregates.	RR §9.8	Nothing in the Figma asks for one. Stays banned; §9.8 row untouched.
P4	No bare percentages. Every base rate ships with N and its Wilson CI; N-gate (suppress <30, no CI 30–99, full ≥100); WEAK cap when the CI spans the always-up baseline.	plan §15.2; <code>components/BaseRate.tsx</code> ; skill <code>base-rate-display</code>	<code>BaseRate</code> remains the sole renderer, restyled only. The card-level proportion bar prints counts (§1.3 #1).
P5	Misses stay first-class; the log stays append-only. No win-only track record.	plan §15.7; DB triggers	Track-record redesign shows hits and misses with equal visual weight — chips differ in hue, never in size or emphasis (§5.4).
P6	No trending/most-bought surfaces; no gamification (streaks, XP, confetti); no urgency theater; no anthropomorphic AI voice.	RR §9.8; plan §15.8	No leaderboards, no pulse animations, no celebratory states anywhere in the new system. Review queue stays timer-free and score-free.
P7	Gain/loss is colorblind-safe and redundantly encoded. A delta never relies on color alone.	plan §3.3; RR §9.7	New pair #2563eb/#ea580c is colorblind-safe; triangles + signs + words remain mandatory. Hollow/filled candles kept. Dot arrays: filled hit vs hollow miss (shape channel added, §3.8).

#	Rule	Lives in (unchanged)	Redesign consequence
P8	Movers need a catalyst or the noise line.	plan §1.5; <code>copy.mover.noNews</code>	Row anatomy keeps the catalyst zone; noise renders muted italic, never hidden.
P9	LLM narrates, never computes; the verification gate blocks unverified numbers.	plan Appendix E	Pure pipeline concern; untouched by the redesign. Amber gate flag keeps its reserved color (P11).
P10	Desk/Academy separated, doorways + return rails; paper-first; login wall.	plan §1.5; <code>proxy.ts</code>	Rooms keep separate shells and backgrounds (D4). Return rails restyle, never disappear.
P11	Amber is reserved. Exactly two consumers: the verification-gate inline flag and the fired-signal watchlist marker.	plan §3.3	Survives the colored-chip amendment: tier-moderate and grade-mixed get their own amber-adjacent hexes so no chip shares the alert token. Grep stays.
P12	Mechanical voice; all copy from <code>copy.ts</code> ; timestamps everywhere; numbers only via <code>BaseRate / format</code> .	CLAUDE.md; plan §3.9	New copy strings (Appendix B) follow the deck's voice. Every surface keeps its as-of stamp.
P13	Position/length over angle/area — no gauges, donuts, speedometers.	RR §9.7 last clause	Kept verbatim; it is an honesty-of-perception rule, not an austerity rule.
P14	One hero figure per view.	plan §3.1 required list	Kept — it is information hierarchy, and it is also what makes the new hero beautiful.

2.3 Why the line sits exactly here

Every amended rule is about **how the product feels**; every preserved rule is about **whether the product can lie**. Shadows, radii, gradients, and motion cannot make a claim; a projected line, a bare percentage, a win-only log, an animated probability, or a color-only delta can. The test applied to each clause was: "*could a beginner be misled if this rule flipped?*" — if yes it stayed, if no it moved to taste and the user's taste now says warm and modern.

The design system

Every value the build needs, as tokens. Appendix A is the paste-ready `globals.css` v2 — this part explains the values; the appendix is the artifact. The token discipline is unchanged: one sheet, `@theme static`, no hex anywhere else (except `lib/tokens.ts` for the OS-facing constants), `/styleguide` renders it all, greps enforce it.

3.1 Type families (four, no more)

Token	Family (next/font)	Weights/styles	Role
<code>--font-display</code>	Playfair Display	600, 700, italic 400	Page titles, card titles, the Desk date hero, pull quotes. Never a number, never body text.
<code>--font-ui</code>	Inter	400, 500, 600	Every label, control, table cell, nav item. Replaces Archivio.
<code>--font-mono</code>	JetBrains Mono	400, 500, 600	Every numeral in the product, plus mastheads, tags, provenance lines. Replaces IBM Plex Mono.
<code>--font-prose</code>	Newsreader	as today (roman+italic)	Academy lesson bodies, briefing paragraphs, and small-size editorial italics (setup pattern names) — Playfair is a display face; Newsreader is the text serif. They do different jobs.

Rules: numerals are mono without exception (unchanged law). Mastheads move from Archivio Expanded to JetBrains Mono uppercase with `tracking-[0.08em]` — the Figma's masthead voice. **Serif floor:** Playfair renders only at `--text-title` (19px) and above, and never italic below the display sizes; editorial italics at card sizes are Newsreader italic (a text serif holds its hairlines at 15–19px; a display serif does not). `lib/fonts.ts` swaps loaders; all four via `next/font/google` (self-hosted at build, no runtime Google requests — PWA/offline unchanged; next/font's automatic metric-adjusted fallbacks stay on so a slow load keeps line geometry). **Playfair loads with `display: "swap"`** — it sets headlines, and a permanently missed `optional` swap would ship Georgia headlines for the session (§7.4). Font budget: raise `scripts/check-font-budget.mjs` limit from 320KB to **560KB**, and R1 prints the actual per-file woff2 sizes into the script output. If the total exceeds budget, the drop order is: Inter 500 (600 covers emphasis), then Playfair 600 (700 covers titles) — **never Playfair italic 400**, which the login headline and pull quotes depend on.

3.2 Type scale (existing scale kept, four additions)

The px scale in `globals.css` survives (11 → 64px, paired line-heights). Additions:

Token	Value	Use
<code>--text-title: 19px / lh 1.3</code>	card/module titles set in <code>--font-display</code> 600	
<code>--text-display: clamp(30px, 4vw, 46px) / lh 1.1</code>	page-level serif headline (Desk date, room titles, login)	
<code>--text-display-hero: clamp(38px, 5vw, 56px) / lh 1.05</code>	login brand panel only	
<code>--text-input-touch: 16px</code>	form controls below <code>md</code> — iOS zooms-on-focus any input under 16px and never zooms back; every <code><input>/<textarea>/<select></code> renders at ≥16px on touch, whatever the surrounding UI scale (§4.1 lists the components this hits)	

The 64px `--text-hero` numeral (48px phone) is untouched — the hero figure law holds. Body UI stays 13.5px/15px; prose stays 17.5px Newsreader (Academy line-height rises to 1.7, §5.6).

3.3 Color

Full literal tables live in Appendix A (light Desk, Midnight Desk, warm Academy). The system:

Neutrals (Desk, light). Paper `#f9f8ff`; wash gradient stops `#f9f8ff → #eef2ff → #f0f4ff → #faf8ff` (145°); ink `#0f0d1a`; ink-2 `#3d3a4f`; muted `#6e6c80` (Figma's `#7c7a8e` darkened one step — it measured ≈3.9:1 on paper; provenance text must clear 4.5:1); faint `#a9a7b8` (placeholders/disabled, non-text ≥3:1 only); hairline `rgba(99, 88, 143, 0.13)`, hairline-strong `rgba(99, 88, 143, 0.24)`.

Accent (interactive only, never on data or status). `--color-accent #6366f1` (washes, icons, marks), `--color-accent-deep #4f46e5` (text/links on light — 4.9:1), `--color-accent-soft rgba(99, 102, 241, 0.10)` (active-pill washes), `--color-accent-muted rgba(99, 102, 241, 0.06)` (tinted panels). `--gradient-brand: linear-gradient(135deg, #4f46e5, #7c3aed)` — exactly two consumers: the logomark tile and primary buttons. Petrol is retired. The market-state dot in the nav is **not** accent (it is status, not an affordance): muted when closed, ink when open, always beside its text label.

Band (uncertainty fills — data, non-directional). `--color-band: #6d648c`, used at tokened alphas for range-band fills, and nothing else. It is deliberately grayer than the accent so an uncertainty band never reads as clickable, and deliberately not the up/down pair so a symmetric historical range never carries directional valence.

Semantic pair (data only). Up `#2563eb` / down `#ea580c` for chart strokes, candle bodies, triangles, ≥21px figures. Text variants at ≤18px: up-text `#1d4ed8` (≈6.3:1), down-text `#c2410c` (≈5.1:1) — Figma's `#ea580c` measured ≈3.6:1 on white, below AA. Soft washes `rgba(37, 99, 235, 0.08)` / `rgba(234, 88, 12, 0.08)` for delta chips. Redundant encoding stays mandatory (triangle + sign + word).

The amber–orange band is reserved as a REGION, not a hex. Its only occupants: the semantic down/loss pair and the two alert consumers (`--color-alert #8a5200`, wash retuned `#f7ead0`). No chip, tier, grade, importance marker, or module hue may sit anywhere in the amber–orange range — reservation is about perceptual salience; a Desk full of amber-ish "moderate" chips would drown the gate flag even with disjoint hexes. This is a stated rule in the styleguide and an eyeball item in §3.10.

Tier chips (colored, AA text on their washes, word always inside). strong `#047857` on `rgba(5,150,105,0.10)`; moderate **teal** `#0f766e` on `rgba(13,148,136,0.10)` (moved out of amber per the rule above — the green→teal→gray ramp is ordinal, which suits an evidence scale); weak `#6e6c80` on `rgba(124,122,142,0.10)`. WEAK-cap logic untouched.

Evidence grades (inside `Tag` only, never without their word). supported `#047857`, mixed `#0f766e` (same reasoning — "middle of the evidence scale" reads consistently across both taxonomies), weak `#a13d22`, folklore `#b91c1c`. Rust/red sit outside the reserved amber–orange band's center and always carry their word.

Midnight Desk (D1). Paper `#14121f`; wash `#14121f` → `#191627` → `#151329`; surface glass `rgba(30,27,48,0.72)`; ink `#ece9f8`; ink-2 `#c3c0d6`; muted `#918ea6`; hairline `rgba(167,157,210,0.16)`; accent `#818cf8`, accent-deep(text) `#a5b4fc`; band `#8f86b8`; up `#60a5fa` / up-text `#93c5fd`; down `#fb923c` / down-text `#fdb474`; alert `#e0a83e` / wash `#3a2f14`; tier-moderate `#2dd4bf`-family (Appendix A); grade dots keep the P6 dark values except mixed → teal. Orb opacity halves. Shadows deepen (`rgba(0,0,0,0.45)`) and every glass surface gains a visible hairline (shadows barely read on dark; the border carries the edge).

Warm Academy (D4). Paper `#fbf6ee`; wash `#fbf6ee` → `#f8efdf` → `#faf3e8`; hairline `rgba(143,110,72,0.15)`; orbs rose `rgba(217,119,87,0.08)` + gold `rgba(217,164,65,0.07)`; surface **solid** `#ffdf8` — the Academy deliberately does NOT use glass (§5.6: it is a reading room, not a control surface). Ink/muted shared with Desk light. Interactive accents stay indigo. Module category hues (decorative only, §5.6): foundations dusk-blue `#4a5a8c`, structure sage `#5b7a4f`, patterns plum `#7a4a6b`, risk rose `#9c4460` — none in the amber–orange band, none equal to the accent.

Contrast contract. Every text token/surface pair above was hand-computed to $\geq 4.5:1$ (or $\geq 3:1$ for $\geq 21\text{px}$ figures). R1 adds a **contrast table** to `/styleguide` (computed live from tokens with a tiny luminance helper) and a unit test that fails if any named pair drops below its threshold — drift-proof, not vibes-proof.

3.4 Surfaces and elevation

Level	Class	Recipe	Used by
L0 wash	<code>.app-wash</code>	fixed-position layer: wash gradient + two orbs (<code>--orb-1/2</code> radial tokens). <code>aria-hidden</code> , <code>pointer-events:none</code> , <code>z-0</code> . Never <code>background-attachment: fixed</code> (iOS).	both room layouts
L1 card	<code>.surface</code>	<code>--color-surface</code> , <code>--radius-card</code> , hairline border. No blur .	every Desk module card
L2 raised	<code>.surface-raised</code>	<code>--color-surface-raised</code> + <code>--shadow-soft</code> + hairline. Blur only $\geq \text{lg}$ via <code>lg:backdrop-blur-xl</code> .	stat cards, hover-emphasized cards
L3 sticky	<code>.surface-bar</code>	<code>--color-surface-bar</code> (paper at 85% alpha) + <code>backdrop-blur-lg</code> + hairline edge	top nav, bottom tab bar
L4 overlay	<code>.surface-overlay</code>	<code>--color-surface-overlay</code> (96% alpha) + blur + <code>--shadow-overlay</code> over <code>--scrim</code> (<code>rgba(15,13,26,0.4)</code>) + 4px scrim	rail, sheet, palette, popover

These five class bodies (and the `@supports not (backdrop-filter: blur(1px))` solid fallback) are **written out in Appendix A** — the executor pastes, never invents. All surface alphas are tokens with dark twins. One Tailwind v4 caveat, stated here so nobody trips on it: `rgb(r g b / a)` color tokens work as utilities (`bg-surface`), but the slash-opacity modifier (`bg-surface/50`) silently no-ops on them — never use it; if a new alpha is needed, mint a token.

Shadow tokens: `--shadow-soft: 0 4px 24px rgba(99,88,143,0.08)`, `0 1px 4px rgba(0,0,0,0.04)`; `--shadow-lift: 0 6px 24px rgba(99,88,143,0.12)`; `--shadow-overlay: 0 8px 40px rgba(99,88,143,0.15)`. Dark variants in Appendix A.

Blur budget (hard rule, grepped): `backdrop-blur` may appear only in the five sanctioned components: top nav, bottom tab bar, RailDialog, CommandPalette, GlossaryPopover. Cards never blur on phones. **iOS radius clipping:** Safari does not clip `backdrop-filter` to `border-radius` on the same element — every rounded blurred surface wraps its blur in an inner element with `overflow:hidden` (the `.surface-bar` / `.surface-overlay` recipes in Appendix A encode this shape). `prefers-reduced-transparency` falls back to solid surfaces **at every theme scope** — the override must be declared for `:root`, `[data-theme="dark"]`, `[data-theme="system"]`, and the Academy shell alike, because a custom property resolves at the nearest defining ancestor and a `:root`-only override loses to the theme blocks (Appendix A encodes this too).

3.5 Shape

`--radius-card: 16px` · `--radius-panel: 12px` (nested tinted panels) · `--radius-control: 10px` (inputs, buttons) · `--radius-chip: 8px` · `--radius-pill: 999px` (nav pills, dots, bars). `--radius-edge` (2px) is **deleted, atomically**: R1 performs a global class rename in the same commit that lands the new sheet (§8 R1) — Tailwind emits no error for a utility whose token vanished; it just stops styling, so orphaned classes are found by grep, not by the compiler. Arbitrary radii stay banned.

3.6 Motion

Tokens: `--duration-quick: 150ms` (hover, focus, chips) · `--duration-fade: 200ms` (reveals, page transitions, chart appearance) · `--duration-slide: 240ms` (rail/sheet) · `--ease-quiet: cubic-bezier(0,0,0.2,1)` (the only easing; no springs, no bounce).

Allowed:

- Hover/focus color shifts (150ms). Hover **lift** (`translateY(-1px)` + `--shadow-lift`) only on surfaces whose subtree contains **no** P2 component — a transform on an ancestor moves the probability visual inside it. P2-containing cards get border/shadow/background hover feedback only, no translation.
- Route transitions: a `template.tsx` in each room shell applies a **200ms opacity-only fade** on segment mount. No `translateY`, no scale — the page appears as one settled sheet; every frame shows every visual complete and unmoving relative to everything else. This is the single sanctioned animation that may contain a P2 subtree, and the reason it may be exactly that it contains no relative motion. Pure CSS, no experimental flags, no library.
- Rail/sheet slides (240ms), palette fade+scale (0.98→1 — contains no P2 content), popover fades, chevron rotations on `<details>` summaries.
- `<details>` content fade — **except** any `<details>` whose subtree contains a P2 component (setup cards do: BaseRate, range panel). Those reveal instantly, and a DOM-level test enforces it (§3.10).
- Chart appearance: charts fade in **as completed wholes** (200ms opacity, once, on initial mount, never on data refresh). **No progressive or directional reveal anywhere** — a left-to-right sweep on a price series reads as momentum. (This deletes the earlier `--duration-draw` concept; there is no draw-in in this system.)

- The single quiet skeleton shimmer (1.6s, low-contrast) on loading placeholders — never on probability/money placeholders, which load as text.
- **The signature micro-interaction — the provenance reveal.** The one interaction that could only be this product: every masthead's as-of stamp is interactive; hover/tap fades in (150ms) the full provenance chain inline — "FRED · prior close · fetched 8:41 PM ET". Motionless by default, P2-safe (it is chrome text, not a probability visual), and it turns the app's honesty obsession into its most-touched gesture. Specified once in `SectionMasthead` (§4.1), inherited by every module.

Forbidden (the P2 carve-out, enforced structurally): `BaseRate`, dot arrays, `CalibrationScatter`, `RangeBands` (né `VolBandPanel`), `QuantileDotplot`, proportion/breadth bars, Brier figures, `StatFigure` deltas — these render complete on first paint. No `transition`/`animation` inside their files, **and no animated or transformed ancestor above them**: every P2 component root carries a `data-p2` attribute, and a unit test (jsdom) walks ancestors of every `[data-p2]` node asserting none carries a `transition/animation/transform` (route-fade `template` exempted by its known class). Numbers never tick, count up, or roll anywhere in the app. Nothing loops. Nothing autoplays. Nothing moves to attract attention. `prefers-reduced-motion` still zeroes every duration globally.

Why a candle chart may fade in while a range band may not even do that: the candle chart is a record; fading in a completed record is showing a photograph. The range band is a claim about uncertainty; any entrance treatment reads as "the forecast is arriving." Records may appear; claims are simply there.

3.7 Iconography

`lucide-react` (already in the Figma's vocabulary), 16px at 1.75px stroke (24px in the bottom tab bar), color `--color-muted` default / `--color-accent-deep` active. Functional icons only: nav tabs, close/expand, check/alert list markers, the Settings gear, the palette hint. Never decorative rows, never emoji, never an icon where a word fits better. New dependency — add to `app/package.json` in R1.

3.8 Charts (the visual highlight)

Shared grammar: hairline grid (dashed 2/4, `--color-hairline`), mono 11px axis ticks, tooltip = L4 mini-panel (hairline border, mono values, 150ms fade), every chart carries its as-of stamp and source line. Specifics:

- **Candle chart** (`ticker/CandleChart`): keeps lightweight-charts, hollow-up/filled-down. Reads the new tokens automatically (`getComputedStyle` hook unchanged). Appears as a completed whole (fade), never sweeps.
- **Sparklines** (`Watchlist`): 1.5px stroke in up/down by 14-day direction + a soft area fill via the tokened `--gradient-spark-up/down` (12% → transparent). The in-mark color is color-only by nature; it is legal **only because** the adjacent price + delta chip carries the triangle/sign/word redundancy — that dependency is stated in the component comment.
- **The Range Ladder** (`ticker/RangeBands.tsx`, NEW — replaces `VolBandPanel`'s list as the primary render; §5.5 owns the page spec). One row per horizon (5d / 10d / 20d, the ≤20d rule unchanged), each row a horizontal band on a **signed-return axis**: outer band = the 80% historical range (`--color-band` at 10%), inner band = the 50% range (at 22%), band edges 1px `--color-hairline-strong` with mono 2xs percentile labels, a 1px ink zero line labeled "0% = last close" (the one strong anchor), and the row's 20-dot quantile dotplot rendered *inside* the outer band — the honesty furniture literally is the texture. Labels reuse the pinned `copy.volband.label` ("In the past, 8 in 10 {h}-day paths from here stayed inside this range") + mandatory `volband.caveat`; N and window printed per row. **No 50th-percentile mark of any kind. No connecting silhouette between horizon rows** — joining the bands into a cone would redraw the forecast-fan the product refuses to be; the ladder of discrete horizons is the honest fan, sliced. Static, `data-p2`.
- **Quantile dotplot** (`QuantileDotplot`, standalone + embedded in the ladder): 20 dots = 20 equal-probability historical outcomes, 9px; above zero = filled `--color-up`, below = hollow `--color-down` with **full-opacity 1.5px stroke** (the shape channel must survive at dot size). Caption: `copy.dotplot.caption`. Static, `data-p2`.
- **Dot array** (inside `BaseRate` detail): every dot = one case (never scaled); hits filled, misses hollow with the same 1.5px full-opacity stroke rule; 9px, 3px gap, wraps; legend with counts. Static, `data-p2`.
- **Proportion bar** — lives **inside** `BaseRate` **only** (it is a base-rate display, so the N-gate, CI, baseline, and WEAK cap must travel with it — no other component may render an "X of N" visual, and §3.10 greps for strays). Equal-visual-weight encoding: both segments at the same alpha (hits `--color-up` 35%, misses `--color-down` 35%), lengths carry the proportion, printed counts carry the numbers — saturation never exaggerates the majority.
- **Calibration scatter** (`CalibrationScatter`): keeps radius=√N; hits/misses as filled/hollow; dashed diagonal labeled "perfect calibration" in mono 2xs. Static, `data-p2`.
- **Breadth strip** (MacroPulse): slim pill bar, advancing/declining segments at **equal alpha** (35%), mono adv/dec counts adjacent. Static, `data-p2`.

3.9 Spacing & layout

4px base grid (unchanged). Card padding 20px (24px ≥lg). Module gap 24px. Page gutters 16px phone / 32px desktop. `max-w-[1360px]` container kept. Desk becomes **two-column at desk: (1366px)** — a broadsheet spread, deliberately: the main column keeps the narrative ritual in order (Pulse full-width, then Brief → Movers → Setups → Scorecard) and the 340px sidebar holds the *reference* modules a reader glances at rather than reads through (Watchlist, Calendar in its compact variant §5.1, SourceStatus). The ritual's strict 1→8 order remains the invariant on phone and in the DOM (grid areas place the sidebar; tab order = ritual order). This is how a front page works: sequence in the main column, standing matter in the rail. Touch targets ≥44×44px on every interactive element.

3.10 Anti-drift checklist v2 (run at every phase exit, replaces plan §3.10 greps)

Mechanical greps — must all be empty outside `app/globals.css` + `lib/tokens.ts` (items marked ⊗ are new since the adversarial review):

1. Hex colors anywhere else (excluding the two sanctioned files, SVG fixtures).
2. Raw gradients: `linear-gradient|radial-gradient` outside `globals.css`.
3. Arbitrary shadows/radii: `shadow-\[|rounded-\[`.
4. `backdrop-blur` outside the five sanctioned components (§3.4).
5. Amber tokens (`alert`) outside the two consumers.
6. Motion on P2 files: `transition|animate|@keyframes` inside `BaseRate.tsx`, `CalibrationScatter.tsx`, `RangeBands.tsx`, `QuantileDotplot.tsx`, `StatFigure.tsx`, `SetupCards.tsx`, `MacroPulse.tsx` (the last two carry

inline P2 marks today) — plus the ☸ **jsdom ancestor test**: no animated/transformed ancestor above any `[data-p2]` node.

7. Font families beyond the four loaded; `small-caps`; ☸ `font-stretch-\l` (dead once Archivo leaves); count-up/ticker libraries; confetti/particle libraries.

8. `background-attachment` anywhere.

9. ☸ Dead legacy classes after the R1 rename: `desk-bg|academy-bg|rounded-edge|font-stretch-\l` (the R1 mapping table, §8) — Tailwind drops unknown utilities silently, so only this grep catches them.

10. ☸ Base-rate strings outside `BaseRate`: `of \{?\d| of (n|N)\b` display patterns / "N=" + percentage pairs rendered anywhere except `BaseRate.tsx` (the sole renderer rule made mechanical).

11. ☸ Slash-opacity on token colors (`bg-surface/`, `border-hairline/` ...) — silently no-ops in Tailwind v4 with `rgb(/)` tokens; must be empty.

Eyeball checks against fresh screenshots (each applies only once its surface exists — R1 gates on the styleguide only; the wash item starts at R2): one hero figure per view; every module masthead numbered + stamped; chips always carry words; misses visible wherever hits are; the wash present but never in front of content; **nothing in the amber-orange region except losses and the two alert consumers**; nothing moves after first paint on a settled page except pointer feedback.

Components

Shared primitives first, then the shell, then overlays. Every spec names the file, what changes, and what must not. "Restyle" always means: markup semantics, server/client split, aria, and tests' assertions on CONTENT stay; only classes/tokens and layout change.

4.1 New shared primitives (R1 builds these; later phases consume them)

components/Surface.tsx — the card, at last a component instead of repeated class strings. Props: `level` (`card` | `raised` | `tinted`), `as` (default `section`), `className` passthrough. Recipes from §3.4 (the class bodies are in Appendix A); `tinted` = `--color=accent-muted` bg + `accent-15%` border + `--radius=panel` (the Figma's base-rate panel). Migration is incremental: each phase converts the surfaces it touches; the grep for leftover `border-hairline` `bg-surface` combos happens at R6, not before.

components/AppWash.tsx — the L0 wash layer: `<div aria-hidden class="app-wash">` with two orb children. **Mounting rule (load-bearing)**: the wash div must render *inside* the element that carries `data-theme` (and inside the Academy shell that remaps the room tokens), as the first child, with all content in a sibling `relative z-10` wrapper — a wash mounted outside the theme scope reads the light `--gradient=wash` forever. Room decides hue via the token scope it sits in; no props needed.

SectionMasthead v2 — anatomy unchanged (index number · title · rule · timestamp). Restyle: title stays mono uppercase in `--color=muted` (the Figma's eyebrow voice — accent on eight mastheads would make indigo mean "chrome"; accent belongs to interactive text and small in-card slot labels only); the 2px ink rule becomes 1px `--color=hairline-strong`; the index number renders in `--color=faint`. **The timestamp is the signature interaction (§3.6)**: rendered as a faint chip; hover/tap fades in the full provenance chain inline ("FRED · prior close · fetched 8:41 PM ET") from a new `provenance` prop. One component, every module inherits the gesture.

StatFigure v2 — hero/figure/body scales unchanged, hero always ink (law). The delta moves into a **soft chip**: mono text in up-text/down-text on the matching wash, `--radius=chip`, triangle + sign kept. Root carries `data-p2`. Law kept and re-asserted in the file comment: semantic color never exceeds `--text=num-lg`.

Tag v2 — variants and word-always rule unchanged; visuals: `--radius=chip`, soft washes per §3.3, 6px round dot replaces the 2px square (pill amendment). `tier` variant now colored (strong green / moderate teal / weak gray tokens); `catalyst` stays neutral (ink on `rgba(99,88,143,0.08)`); `folklore` uses grade-folklore on an 8% wash of itself.

Buttons (recipe block in `/styleguide`, still inlined per surface — a Button component is NOT required, matching current architecture): primary = `--gradient=brand` white text, `--radius=control`, hover brightness 1.05 + `--shadow=lift`; secondary = surface + hairline border, hover border-strong; destructive-quiet = down-text text + hairline; all ≥44px tall on touch, `disabled:opacity-60`. **Exception (unchanged product rule)**: in the cooling-off interstitial, "Sit with it" takes the primary style and "Proceed" the secondary — friction is the point.

Inputs — surface bg at 70% alpha, hairline border, `--radius=control`, 12px/14px padding; focus: border `--color=accent` + `0 0 0 3px rgba(99,102,241,0.12)` ring (replaces petrol outline app-wide via the `:focus-visible` global). Labels stay mono 2xs uppercase muted. **Touch rule (§3.2)**: below `md`, every `<input>/<textarea>/<select>` gets `--text=input-touch` (16px) — iOS zooms on any focused control under 16px and stays zoomed. Components this hits (update each): `LoginForm`, `PaperEntryForm` (all fields + selects), `JournalPrompt`, `AddWatchListForm`, `ForecastResolver`, `CommandPalette` input, `PaperLedger` close form.

OfflineRibbon / **SourceStatusFooter** / **ExternalLink** — content and behavior untouched; ribbon becomes a neutral L1 strip (never amber — P11), footer keeps FRED attribution, links adopt accent-deep underline-on-hover.

4.2 Navigation

The z-ladder (global, load-bearing — every overlay and bar obeys it):

Layer	z	Notes
<code>.app-wash</code>	0	fixed, pointer-events none
content wrapper	10	<code>relative</code>
top bar + bottom tab bar	30	<code>.surface-bar</code>
overlay scrim (rail, palette, alertDialog)	40	<code>--scrim</code> + 4px blur
overlay content (rail/sheet/palette/popover/dialog)	50	<code>.surface-overlay</code>

When any overlay is open, the tab bar sits *beneath* the scrim (30 < 40): dimmed, and `pointer-events:none` while `[data=overlay=open]` is set on the shell (belt and braces — Radix already inerts the background tree). The open overlay owns the bottom safe-area inset; `main`'s tab-bar padding is irrelevant behind the scrim, so insets never double up.

Desktop top bar (all viewports ≥md). File: `(desk)/layout.tsx` `DeskNav` + `components/desk/RoomNav.tsx`. Keep: single row, left cluster brand + rooms, computed active state from `lib/nav.ts` (pure logic untouched, tests untouched), `aria-current="page"`. **Newly sticky** (today's bar is static — this is an upgrade, not a keep): `sticky top-0` on the `.surface-bar`; iOS Safari can break `position:sticky` under an ancestor with `overflow-x:hidden`, so R2 verifies on device and falls back to `fixed` + content top-padding if needed (the html-level overflow guard stays either way; log the outcome). Restyle: brand = 28px `--gradient=brand` tile with mono white letter + wordmark in mono 12px uppercase `tracking-[0.12em]` accent-deep (D3); room links = Inter 13.5px sentence-case, `--radius=control` pills, active = `--color=accent-soft` bg + accent-deep 600, hover = ink-2 → accent-deep 150ms. Right cluster: market-state static dot (muted closed / ink open — status is not an affordance, §3.3) + its text label + mono date + **⌘K hint chip** (mono 2xs "⌘K" in a faint-bordered chip; click opens the

palette) — hidden <md. The Academy top bar keeps its separate minimal shell restyled with the same recipes over the warm wash; **on phone (<md) it drops the "← Back to Desk" text link** (the bottom bar's Desk tab is the return path; two doorways on a 375px row is one too many) and keeps it ≥md.

Phone (<md), D2 recommended: bottom tab bar. New `components/desk/TabBar.tsx`, mounted in both room layouts (client component reading `usePathname` → `activeRoomHref`). Fixed bottom, z-30, `.surface-bar` glass, top hairline, `padding-bottom: env(safe-area-inset-bottom)` and `padding-left/right: env(safe-area-inset-left/right)` (landscape notches — browser tabs are not orientation-locked even though the installed app is); five tabs — Desk (`Newspaper`), Scans (`ScanSearch`), Paper (`NotebookPen`), Track (`LineChart`), Academy (`GraduationCap`) — 24px icons + 10px Inter labels, min 44px targets, active = accent-deep icon+label + 3px pill indicator dot above the icon. Settings leaves the bar: a `Settings` gear icon sits right in the phone top bar. Main content gets `pb-[calc(64px+env(safe-area-inset-bottom))]` in both rooms. The bar never hides on scroll (calm; position is trust) — **except for the keyboard:** iOS does not resize the layout viewport when the soft keyboard opens, so a `bottom:0` bar floats over the content mid-screen; the TabBar listens to `visualViewport` resize and hides itself (`hidden`, not animated) while `visualViewport.height` is materially below `innerHeight` (~150px threshold), restoring on keyboard dismissal. Tab switches reset scroll to top (App Router default — stated as intended: each room is re-entered at its head, the ritual's start). `e2e/nav.spec.ts` extends: phone project asserts tab bar visible, tabs navigate, active tab tracks route, bar hides on input focus, no horizontal page scroll (kept assertion).

Phone Plan B (if D2 vetoed): RoomNav strip stays but gains: `scroll-snap-type: x proximity`, auto-`scrollIntoView({inline:'center'})` of the active link on route change, 16px edge fade masks (`mask-image` gradient) signalling overflow, wordmark shrinks to the mark tile only (<md), Settings pinned last with a leading hairline. Same restyle tokens.

CommandPalette — behavior and index mechanics unchanged; the results get a **typed hierarchy** so the palette teaches the product's map: three labeled groups in fixed order — Rooms (icon + name), Tickers (symbol mono + last price mono-muted), Lessons (title + read-tick ✓) — each group under a mono 2xs uppercase muted header, zone-chipped Desk/Academy as today. Restyle: L4 overlay, 12px radius, `Search` icon input row, fade+scale 200ms in (contains no P2 content), none out. Phone: opens from the top-bar search icon; panel anchors to the top (`pt-[12vh]` as today) and caps its height to `visualViewport.height` minus the keyboard so the active row never hides behind it; input at `--text-input-touch`.

4.3 Overlays

RailDialog — Radix behavior/history handling untouched. Desktop: 440px right rail as L4 (z-50 over its z-40 scrim), 240ms translateX slide-in. Phone: bottom sheet with 16px top radii, **grab handle** (32×4px `--color-faint` pill), `padding-bottom: env(safe-area-inset-bottom)` (it owns the inset while open — the tab bar is behind the scrim), 240ms translateY. Blur wrapped per the iOS clipping rule (§3.4). Reduced-motion: appears in place.

GlossaryPopover — L4 mini-panel, `--radius-panel`, 150ms fade+2px rise; dotted underline term becomes accent-deep dotted. Review-queue seeding behavior untouched.

Cooling-off interstitial (PaperEntryForm) — `role=alertdialog` unchanged; renders on `--color-surface-solid` (a decision moment: **no transparency, no gradient**), hairline border, z-50 over its own z-40 scrim, the two buttons per §4.1's exception. No entrance animation (it is friction, not delight).

WorkedExampleDrawer — the schematic stays schematic (P1; never real prices), but the reveal becomes **staged pedagogy** rather than a dump: the drawer opens with the bare path + the "not X's prices" caption (150ms fade — the SVG path itself never animates; it *looks* like a price path and must not *behave* like one); each of the three steps, on hover/focus, lights its numbered marker in accent AND sets the step's one-sentence "what to notice" line under the chart (existing content, resequenced) — the reader walks the pattern marker by marker instead of parsing a legend. Markers restyle to accent; keyboard path: steps are focusable in order, `aria-describedby` ties marker to step.

Pages, room by room

Page order = build order within each phase. Every page keeps its data flow, server actions, copy keys, and honesty behavior; what changes is named per page. Phone behavior is stated inline, not deferred to a "mobile section" that Opus might read too late.

5.1 The Desk (/)

Header (new). Above module 01: mono 2xs uppercase muted eyebrow "THE DESK — EVENING EDITION" (from `copy.desk.edition`; our edition is the 9pm briefing, not the Figma's morning fiction), serif `--text-display` date ("Friday, July 11, 2026" — the run date of the data), and a muted status line built from existing data: market open/closed + "data as of Fri close · updated 8:41 PM ET" (timestamps everywhere; the line doubles as the as-of context for every delta chip below it). No countdowns (urgency).

Layout. Phone/tablet: the single ritual column, order unchanged (heartbeat → 01 Pulse → 02 Brief → 03 Calendar → 04 Movers → 05 Watchlist → 06 Setups → 07/08 placeholders → Scorecard → SourceStatus). At `desk`: 1366px: the broadsheet spread (§3.9) — CSS grid `minmax(0,1fr) 340px`, gap 24px; Pulse spans both columns; the sidebar takes the reference modules (Watchlist, Calendar **compact variant** below, SourceStatus); DOM order stays ritual. Every module becomes a `Surface level=card`; placeholders keep their masthead + one muted line + quiet shimmer.

Module specs.

- **01 Macro Pulse** — content per Part 6 fix (true index levels). Hero: S&P 500 level 64px ink + delta chip; row of body StatFigures: Nasdaq Composite, Dow, VIX, 10-yr (each with a delta chip where a prior level exists — the delta is prior-close-over-previous-close, the same as-of as the level; the header status line and the masthead stamp carry that context on-surface, and the chip's `aria-label` says "1-day change to Friday's close"); small-caps slot renders the IWM ETF proxy **with its proxy chip** (§6.1); breadth strip per §3.8. Provenance line under the grid: `copy.macro.provenance` + as-of stamp (and the masthead's provenance reveal carries the chain). Phone: hero full-width, figures wrap 2-up.
- **02 Brief** — serif `--text-title` headline (Newsreader italic stays for "Today's focus"), labeled slots keep their mono 2xs slot mastheads in **accent-deep** (the Figma's one sanctioned in-card accent use), hairlines between slots, superscript source links in accent-deep. **Held state (designed, not just worded)**: the card renders the full slot skeleton — all four slot mastheads in `--color-faint` with quiet placeholder rules where prose would sit — and `copy.brief.unavailable` once beneath the first. The structure still teaches what a briefing contains even when tonight's is held. Never amber.
- **03 Calendar** — per Part 6 fix. Full-width row (main column, phone): date cell (mono, two lines: "TUE JUL 15" / time or "—"), the **code chip** (`Tag` catalyst variant showing the allowlist code: CPI, JOBS, PPI, GDP, PCE, RETAIL, FOMC, EARNINGS — one vocabulary, Appendix C is its single source), title (symbol bolded), consensus/prior mono cell when present. High importance = a 6px ink dot + the word "high" in mono 2xs beside the title (never amber, never red, never the loudest thing in the row — P6/P11). **Compact variant (340px sidebar)**: two lines — line 1 = date + code chip + importance dot+word; line 2 = title truncated; consensus/prior drop to the row's rail drill. **Empty state is a signature, not an apology** (§6.2 makes it likely): a full-height quiet card — centered serif line `copy.calendar.empty` ("A quiet stretch — no scheduled catalysts in the next 14 days."), hairline rule, mono sub-line `copy.calendar.emptySub` ("Curated from the full FRED release feed; the rest was noise."). The absence of noise is the product working, and the card says so.
- **04 Movers** — adds the last price under the symbol (mono 2xs muted); change% chip in up/down-text on wash; RVOL emphasized `--color-accent-deep` when $\geq 2\times$; catalyst zone unchanged (chip + headline + ExternalLink, or the noise line in muted *italic*). Rows stay RailTriggers with `hover:bg` shift to accent-muted (background only — rows contain delta chips, so no transform, §3.6). Empty state: `copy.mover.quiet` ("No moves cleared the catalyst-or-noise bar today.") in the same quiet-card voice as the calendar. The RelVol footnote enters `copy.ts` (Appendix B — reworded in review; no "not yet public" FOMO bait).
- **05 Watchlist** — sparkline per §3.8 (stroke + tokened soft fill, 36px tall), price + delta chip right-aligned, reason line muted, fired-signal marker keeps reserved amber (P11).
- **06 Setup cards** — collapsed summary row: pattern name in **Newsreader italic** `--text-title` (§3.1 serif floor: Playfair is display-only), symbol mono, colored tier chip, and **n=108 in mono muted — nothing else numeric**. No proportion, no percentage, no bar on the collapsed card: a base rate may not appear without its CI, baseline, and WEAK-cap context, and those live in `BaseRate` (P4 — this was the honesty review's top finding, fixed by moving the proportion bar inside `BaseRate`). The `<details>` expansion is **instant** (its subtree is P2 — §3.6); only the chevron rotates. Expanded: `BaseRate` in a `Surface tinted` panel (its restyle adds the equal-weight proportion bar + dot array under the sentence, all `data-p2`); typical-range panel as a **flat neutral band** (`--color-band` at 14%, no gradient — a gradient across a range implies density and valence the interval doesn't claim) with 25th/75th mono labels and the pinned "historical distribution, not a price target" copy; "What fired" list with `Check` icons in accent; weakeners with `AlertCircle` in down-text; `WorkedExampleDrawer` (§4.3); scope line; doorway link.
- **Scorecard + Journal** — Surface card, serif title, hits/misses in outcome chips (§5.4), journal textarea per §4.1 inputs (16px on touch), Brier anchor line unchanged.

5.2 Scans (/scans) — the visible recipe, staged like one

The scan page's identity IS the product's anti-black-box stance: the recipe is public. Design it as a recipe card, not a text dump. Each preset = `Surface card`: header row with serif `--text-title` name (GlossaryTerm-wrapped where it is one) + grade/FOLKLORE `Tag`; then the criteria as a **numbered clause list** — each criterion on its own hairline-separated row with a mono index (01, 02, 03...) and the clause in UI type at 62ch — a recipe you can check off by eye, not a paragraph you parse; then the results line: "N matches today" as a mono figure + as-of stamp; then matched tickers as a wrapping row of neutral mono chips (≥ 44 px touch targets, each links to `/ticker/[symbol]`), "+ N more" in faint. No percentages on this page — unchanged rule (a match is not a base rate), restated in the page header prose. Empty preset state: "0 matches today" stays visible with the recipe — a scan that fires nothing is information. Phone: cards stack; chips wrap.

5.3 Paper desk (/paper) — the receipt is a receipt

Form in a Surface card (§4.1 inputs — every control ≥ 16 px on touch, 2-col $\geq md$, stacked on phone). **Cost mirror — the app's most honest artifact gets the full treatment**: a `Surface tinted` panel styled as a register receipt — 2px dashed hairline top and bottom edges (the tape), each factor line right-aligned mono in

`--color-muted` (round-trips/yr · avg round-trip bps · annual drag bps), a `--color-hairline-strong` rule, then the total set at `--text-num-lg` in `--color-down-text` with the word "drag" beside it (redundant encoding): "-4.8% / yr drag". The punchline lands with the weight of a bill, because it is one. (P&L-never-hero rule untouched: this is a cost, on /paper, at num-lg not hero.) Frequency-mirror warning stays plain prose (not amber). Half-Kelly helper: tinted panel + figures via StatFigure body scale. Ledger: Surface card table, realized P&L in outcome-style chips (up/down-text + sign + word "gain"/"loss"), open positions with their close form inline. M3 soft-gate aside: hairline card with Academy doorway (never a lock — unchanged).

5.4 Track record (/track-record)

Stats row: 4-5 Surface raised stat cards: mono `--text-num-lg` value + mono 2xs uppercase label (Resolved · Hits · Misses · Unresolvable · Hit rate — hit rate keeps N and misses adjacent; never a lone win number, P5). These cards contain StatFigure's → **no hover transform** (§3.6); they are display, not controls. Table: header hairline-strong, rows hairline, outcome column = chips: hit up-text on up-wash, miss down-text on down-wash, word always inside, equal size/weight (misses are never smaller, P5). Filter chips above the table (all/hit/miss/tier, client-side). "Your forecasts": Brier stat card + restyled CalibrationScatter (§3.8) + ForecastResolver buttons per §4.1. Phone: stats 2-up grid; table keeps overflow-x-auto with a right-edge fade mask signalling scroll.

5.5 Ticker (/ticker/[symbol]) — the Range Ladder ships here

This page's visual hero is the **Range Ladder** (§3.8) — not a number. The price StatFigure renders at `--text-num-lg` (not hero-64) with symbol mono + name serif above it as an eyebrow; the hero-figure law is satisfied (no 64px numeral on this route) and the ladder takes the stage. Page order: header (symbol · name · price + delta chip + as-of) → candle chart in a Surface card (tokens re-read automatically; fades in complete) → **RangeBands** (replaces VolBandPanel's list as the primary render; the list stays beneath as the accessible table/text equivalent — screen readers and the no-SVG fallback read the same numbers): one row per horizon, nested 50/80 bands in `--color-band`, embedded quantile dotplots, ink zero line "0% = last close", per-row volband.label + volband.caveat + N + window, no median mark, no cone silhouette (§3.8). WEAK/regime-break state renders the caveat above the ladder, never hides it. Polish budget: perfect band-edge typography with leader lines, its own /styleguide section, dedicated VRT shots both themes. The geometry is a pure function (TDD, R4); the SVG renders the computed geometry.

5.6 Academy — a reading room, not a tinted dashboard (D4)

The room switch must be felt in the furniture, not just the wallpaper. Academy diverges from the Desk in four deliberate ways: **(1) solid warm paper cards** — `--color-academy-surface #ffffd8`, hairline border, NO glass, NO blur, NO shadow (a reading room is not a control surface; the Desk's material says "instrument", the Academy's says "book"); **(2) serif kickers instead of terminal mastheads** — module and section headers drop the numbered mono masthead for a Newsreader italic kicker over a hairline (the index numbers stay in the curriculum map where sequence matters, as plain mono prefixes); **(3) reading typography** — prose lh rises to 1.7, measure stays 65ch, lesson H1 in Playfair display; **(4) warm light** — cream wash + rose-gold orbs (§3.3). Interactive accents stay indigo everywhere (links mean one thing in both rooms).

Home: module groups as solid cards with a 4px category bar per module — **decorative only** (the module title beside it carries the information; the bar is never the sole encoding), hues from the module palette (§3.3, none amber, none accent). Lesson rows with read-tick (Check in accent) + mono minutes, M3 gate note unchanged. The closing pull-quote card renders copy.academy.quote — one fixed quote, never rotating (rotation is a slot machine). **Lesson:** serif display H1, prose measure 65ch Newsreader, mdxComponents restyle only (h2 serif `--text-title`, links accent-deep), retrieval questions in a solid tinted card, return rails top+bottom (top link hidden <md per §4.2), LessonReadBeacon untouched. **Review queue (a calm ritual, designed as one):** one centered solid card at a readable measure, generous whitespace above and below; the term set in Newsreader italic `--text-lg`; "3 of 12" as a plain mono position line (position, not progress-bar gamification); Reveal as a secondary button; after reveal, the definition in prose type + optional "Full lesson →" doorway; "I knew this / Not yet / Skip" as three **equal-weight** secondary buttons (no primary — the UI sells no right answer); 200ms fade between cards; no timers, no streaks, no scores (P6). **Glossary: the term is the hero** — each entry: term in Playfair `--text-title`, part-of-speech/context in mono 2xs muted, definition in prose type, doorway link; two-column masonry ≥lg, hairline-topped entries; alphabet rail (mono letters, anchor links) on desktop.

5.7 Login (/login)

Split screen ≥lg: left brand panel on `--gradient-brand` (150°) with the faint 48px grid-line overlay (Figma), logomark + wordmark (D3), serif `--text-display-hero` headline "Your personal *broadsheet* for the market." (Playfair italic 400 — the token the font-budget ladder protects, §3.1), the licensing/product explanation re-set in Inter on indigo-200, and the Templeton quote block (copy.login.*) — mechanical voice, quote attributed). Right: the form at max-w-[400px] per §4.1 inputs (16px on touch), primary gradient submit "Open my desk →", the "personal tool, no advice" line kept. Phone: brand panel collapses to logomark + one headline above the form. Constraint honored: page stays force-static (SW precache) — all decoration is CSS, no images. auth.spec.ts assertions (wall text, error equality, cookie) unchanged.

5.8 Settings, Offline, Styleguide

Settings: Surface cards per section; watchlist rows per §4.1. The theme control is labeled "Desk theme" — segmented pill (System / Light / Dark), active segment raised — with helper text stating its scope plainly: "Applies to the Desk. The Academy is always light." (D1/D4 made legible at the control, not in a doc). Same form posts, no client JS added. **Offline:** neutral Surface card + masthead, unchanged copy. **Styleguide:** re-organized to the new system — tokens (with the live contrast table §3.3), surfaces, type specimens (all four families incl. the serif floor demo), chips/tags, buttons/inputs, chart specimens (range ladder, dotplot, dot array, proportion-in-BaseRate, calibration, sparkline), motion demos (with a "probability visuals do not move" section showing the static list and the data-p2 convention), the amber-region reservation swatch row, both themes side by side. The styleguide is the VRT anchor (Part 9).

Content fixes (Phase R0 — build FIRST, before any styling)

Two places the product currently tells a beginner something false or noisy. They outrank styling; they ship as their own phase with their own gate. Both fixes are specified to the line because the recon traced them to the line — and re-verified after the adversarial pass.

6.1 Macro Pulse: real index levels, honest proxies (Bug 1)

The defect. `app/lib/morning.ts:40–46` maps ETF tickers to index names (`SPY` → "S&P 500") and `quoteFromCloses` (`morning.ts:72–78`) renders the ETF's close as the labeled value — so the Desk hero says "S&P 500: 754.94" when the index is ~6,800. Same for Nasdaq/Dow/Russell. The day-change % is roughly right (an ETF tracks its index's percentage move); the *level* is mislabeled. A beginner must never read a false number — this is the worst bug in the product regardless of styling.

The fix (chosen: true levels from FRED, ETF fallback that is always labeled honestly). FRED is already integrated with a generic `latest_value(series_id)` (`pipeline/adapters/fred.py:45–69`) and the macro read already pulls `VIXCLS / DGS10` this exact way (`pipeline/jobs/job_a.py:47–48, 98–113`).

- Pipeline (TDD first):** add `latest_two(series_id)` to the fred adapter (same observations call; returns the last two real values) and extend `_read_macro` to fetch series `SP500` (S&P 500), `NASDAQCOM` (Nasdaq Composite), `DJIA` (Dow) — level AND prior level each. **Russell 2000 has no free FRED daily series — the small-caps slot stays an ETF proxy, honestly labeled** (Appendix E-1). Record fixtures `sp500.json`, `nasdaqcom.json`, `djia.json` via the existing `MSM_FIXTURES=1` flow. Failure mode: any series unavailable → `None` → downstream renders the ETF-proxy path for that slot (never a silent mislabel).
- Schema:** `market_context` gains six nullable Decimal columns — `sp500`, `sp500Prior`, `nasdaqComposite`, `nasdaqCompositePrior`, `djia`, `djiaPrior` (level + prior per index; the app can only diff what is persisted — `buildMacro` never sees pipeline observations). Migration: `npx prisma migrate dev --name add_index_levels` (the `--name` flag is mandatory — a bare `migrate dev` prompts interactively and stalls an unattended run). Extend `_upsert_market_context` (`pipeline/publish.py:408–429`) and the `read_macro` return type (`pipeline/nightly.py:78`).
- Seeds:** `prisma/seed.mjs` gains realistic index values (S&P ≈ 6,812.34 + prior, Nasdaq ≈ 22,3xx, Dow ≈ 44,xxx) in its `marketContext.upsert` — without this the seeded e2e desk falls back to the ETF proxy and the R0 gate assertion can never pass.
- App (TDD first, update `morning.test.ts` expectations):** `buildMacro` (`morning.ts:88–118`) becomes source-typed: `type IndexQuote = { label; value; deltaPct; direction; source: "index" | "etf-proxy"; proxySymbol? }.` Index slots prefer the `market_context` level; delta computes from level vs prior level via `lib/format` (`signedPercent / directionOf`); if the prior is null, the delta renders "—" (never borrowed from the ETF). If the level is null, the slot falls back to its ETF close with `source: "etf-proxy"`. **The label may claim an index name only when `source === "index"` — a unit test asserts this coupling** (the regression lock for the whole bug). Fallback labels are honest per-slot: the S&P and Dow proxies are "S&P 500 · SPY (ETF proxy)" / "Dow · DIA (ETF proxy)"; the Nasdaq slot's proxy is "Nasdaq-100 · QQQ (ETF proxy)" — QQQ tracks the 100, not the Composite, and the label must not claim otherwise.
- UI:** `MacroPulse.tsx` renders per §5.1: index slots show level + delta chip (as-of coupling per §5.1); proxy slots show the proxy chip (`copy_macro.proxyChip`) and the ETF price; provenance line `copy_macro.provenance` + as-of stamp + masthead provenance reveal. If ALL index series fail, the whole strip renders proxy-labeled ETF rows — degraded, still honest, and `SourceStatusFooter` already reports the FRED outage.
- Copy (Appendix B):** `macro.provenance`, `macro.proxyChip`, `macro.proxyNote`, `macro.indexUnavailable`.
- Tests:** pipeline `test_job_a` (new series read + None-tolerance + `latest_two`), publish upsert test, `morning.test.ts` (label-source coupling, per-slot fallback labels, delta from level/prior, "—" on missing prior), `MacroPulse.test.tsx` (proxy chip on fallback; hero level formats via `price()`), e2e `desk.spec.ts` seeded assertion updates (SPX ≈ 6,8xx seeded level, proxy chip on small-caps row).

Note on timing: FRED index series update at the close with a lag; the strip is as-of prior close — identical freshness contract to today's EOD product, now stated on-surface by the provenance line and the Desk header's status line.

6.2 Session Calendar: catalysts only (Bug 2)

The defect. `pipeline/catalyst_ingest.py:75–82` ingests **every** FRED release in the 14-day window; `fred.release_calendar` (`adapters/fred.py:71–96`) calls `/releases/dates` with `include_release_dates_with_no_data=true`, which returns daily-noise releases (Coinbase Cryptocurrencies, Commercial Paper, Euro Short Term Rate, Dow Jones Averages...) and repeats releases across dates (three "FOMC Press Release" rows in the fixture). `loadCalendar` (`morning.ts:398–410`) then takes the 15 soonest rows unfiltered. The calendar the product exists to curate is currently a firehose.

The fix (filter at the write path, on the typed adapter rows — the read stays dumb).

- Adapter:** set `include_release_dates_with_no_data` to `"false"` (kills the no-data repetition at the source). Keep limit/sort.
- Allowlist (new `pipeline/catalyst_allowlist.py`, TDD first):** one constant mapping FRED release name → `{code, display, kind, importance}` (Appendix C is the single source of the vocabulary — codes CPI/JOBS/PPI/GDP/PCE/RETAIL/FOMC; the list only ever grows by a deliberate commit). Matching by normalized name containment, with each entry's `release_id` recorded beside it as documentation; the build verifies each name once against the live endpoint and logs any mismatch to QUESTIONS as `[VETO?]`.
- Filtering + dedup happen on the `List[ReleaseDate]`** (the typed adapter rows, which carry `release_id` — the event dicts do not): drop non-allowlisted releases *before* mapping (the current loop appends unconditionally at `catalyst_ingest.py:77–78`; restructure so drops are skipped, never appended as `None`), then collapse duplicates by `(release_id, date)`. Remaining multi-date rows for one release (FOMC press releases around a meeting) are real distinct dates and stay. `_fred_event` then maps an allowlisted row using the table's `kind` (`fed` for FOMC, `macro` otherwise), `title` = display name, `importance` from the table — and a new `code` field.
- Row shape:** `CalendarRow` (app) and the event dict (pipeline) gain `code` and `importance`; the chip renders `code` (§5.1 — never the raw `kind`), FMP earnings rows get `code: "EARNINGS"`. Earnings importance: **"high" when the symbol is in the served core** (`_CORE_SERVED`, a pipeline constant the ingest can see), **else "medium"** — the user-watchlist half of the earlier heuristic is dropped: the watchlist is app-DB state the pipeline doesn't hold at ingest time (Appendix E-4 updated).

5. **Fixture upgrade:** extend `fixtures/fred/release_dates.json` with a realistic CPI + Employment Situation entry (dated inside the window) so the INCLUSION path is tested, not just exclusion — an allowlist tested only on noise proves nothing about recall.
6. **UI:** `CalendarTimeline.tsx` renders per §5.1 (code chip, ink importance dot + word, compact sidebar variant). **Empty state:** new keys `copy.calendar.empty` + `copy.calendar.emptySub` (Appendix B) rendered as the §5.1 signature quiet card. The existing `copy.calendar.noEdge` ("No clear edge either way — that is a valid outcome.") is a *different sentence for a different situation* (an event with no directional read) and is left untouched — the earlier draft of this plan misquoted it; the adversarial pass caught it.
7. **Tests:** `test_catalyst_ingest.py` (allowlist in/out on ReleaseDate rows, dedup, importance + code mapping, no `None` rows), `test_fred.py` (flag change), new `test_catalyst_allowlist.py`, `morning.test.ts` + `CalendarSourceModules.test.tsx` (code chip, importance render, compact variant), e2e seeded calendar assertion update.

Order note: R0 ships both fixes + tests + a green pipeline run on fixtures before any token work begins. The gate (Part 8) requires the two regression locks (label-source coupling; allowlist in/out) to exist and pass.

Mobile & PWA (first-class, not an afterthought)

The phone is where the current UI is weakest (jumbled Scans, off-screen nav, no safe areas). This part is the phone contract; §4/§5 already state per-surface phone behavior — this part adds the cross-cutting mechanics and the PWA chrome. The adversarial pass added an explicit iOS-Safari clause to nearly every item: the automated matrix is Chromium/Pixel-7, and every one of the worst phone bugs found in review is iOS-specific and invisible to it.

7.1 Breakpoints & ergonomics

- Tailwind defaults (sm 640 / md 768 / lg 1024) + the existing `desk: 1366px` variant. Phone = <md throughout this plan.
- **Touch targets $\geq 44 \times 44$ px** everywhere: nav tabs, movers/watchlist rows, chips that act (ticker chips on Scans), `<details>` summaries, form controls, palette rows. R6 runs an automated sweep (Playwright evaluates bounding boxes on the phone project for every `a`, `button`, `[role=button]`, `summary`).
- **Inputs ≥ 16 px on touch** (`--text-input-touch`, §3.2/§4.1) — iOS zooms on focus below 16px and stays zoomed; there is deliberately no `maximum-scale` lock (pinch-zoom is an accessibility right), so the fix is font size, not viewport clamping.
- **Hero scale** drops to `--text-hero-mobile` (48px) — existing mechanism kept.
- **No horizontal page scroll, ever** — the `html { overflow-x: hidden }` guard and the `nav.spec.ts` phone assertion stay; wide tables scroll inside their own container with a right-edge fade mask as the affordance (track record, ledger).
- **Scroll policy stated:** tab/room switches land at the top of the destination (App Router default, intended — each room re-enters at its head). `overscroll-behavior-y: contain` on the scroll root suppresses pull-to-refresh reloads in standalone Android; iOS rubber-band reveals `html`'s `background-color`, which tracks the resolved theme's paper token, so the bounce never flashes a wrong plane.
- Hover-dependent affordances must have a non-hover equivalent: every action is a visible control (unchanged practice, restated because the new system adds hover polish). The provenance reveal (§3.6) opens on tap on touch devices.

7.2 Safe areas & the keyboard (currently enabled but unapplied — fix in R2)

`viewportFit: "cover"` is already set; the insets are not consumed anywhere. Apply:

- Top bars (both rooms): `padding-top: env(safe-area-inset-top)`.
- Bottom tab bar: `padding-bottom: env(safe-area-inset-bottom)` and `padding-left/right: env(safe-area-inset-left/right)` — browser tabs rotate even though the installed app is portrait-locked, and landscape notches eat the outer tabs (D2).
- RailDialog bottom sheet / palette / alerdialog: each owns the bottom inset while open; the tab bar sits behind their scrim per the z-ladder (§4.2), so insets never stack.
- Main content bottom padding accounts for the tab bar height + inset (§4.2).
- **Keyboard:** the tab bar hides (not animates) while the soft keyboard is up, via a `visualViewport` height listener (§4.2) — iOS keeps `bottom:0` fixed elements pinned to the layout viewport, which the keyboard covers, so an unhidden bar floats mid-screen over the focused form. The palette caps its height to `visualViewport.height`.
- Verification: Playwright asserts the computed padding includes the `env()` expression and the bar hides on input focus (phone project); the iOS-specific behaviors (keyboard float, sticky-under-overflow, blur corner clipping) get a **manual iPhone checklist** run at R2 and R6 gates, photographed into `docs/redesign-evidence/` — stated plainly: these cannot be automated in the current Chromium-only matrix.

7.3 PWA chrome

- **Manifest** (`app/manifest.ts`): `background_color / theme_color` → new light paper `#f9f8ff` (install-time frozen light value — unchanged policy; consequence stated: a Midnight-Desk user sees one light splash frame at cold launch. Accepted and documented — the alternative is per-theme install prompts we don't control).
- **Status bar / theme-color** — **must follow the chosen theme, not the OS**. Today the Desk layout keys `viewport.themeColor` on `prefers-color-scheme`, but the page theme comes from the **cookie**; with the new high-contrast palettes any mismatch (OS light + Desk dark, or the reverse) paints a glaring wrong-colored system strip. Fix in R2: the Desk layout already reads the theme cookie server-side — when the cookie is `light` or `dark`, emit a single explicit `themeColor` (`#f9f8ff / #14121f`); only when it is `system` keep the `prefers-color-scheme` media pair. The Academy layout gains its own `export const viewport = { themeColor: ACADEMY_BG }` (single value — the room has no dark mode; today it silently inherits the Desk root's and can go dark over cream). `lib/tokens.ts` constants: `DESK_BG "#f9f8ff"`, `DESK_BG_DARK "#14121f"`, `ACADEMY_BG "#fbf6ee"`. e2e: with a `dark` cookie under an emulated light OS, the rendered `theme-color` meta equals `DESK_BG_DARK`; the Academy route reports `ACADEMY_BG` under both OS schemes.
- **Icons:** regenerate the set (192, 512, maskable-512, monochrome-96) from a new source SVG: `--gradient-brand` rounded-16px tile + white mono letter (D3). Keep filenames/paths so the manifest and precache don't churn. Maskable: 20% safe zone.
- **Service worker:** behavior untouched (precache `/login` + `/offline`, cacheGuard, quiet update line). R1's gate prints a **line-item precache delta**: per-font woff2 sizes (all four families), app-shell delta, static routes — the "<1MB growth" claim is verified as a table, not asserted as a round number (`pwa.spec.ts` keeps its two-static-routes assertion).
- **Splash/standalone polish:** `display: standalone` + portrait kept. The wash renders edge-to-edge behind the safe areas (the `.app-wash` layer is `inset: 0` fixed).

7.4 Performance budget (the glass tax, paid deliberately)

- Blur budget per §3.4 (five components, never cards on phone; iOS corner-clip wrapper rule).
- Orbs are two fixed radial-gradient layers — cheap; they carry `transform: translateZ(0)` to stay on the compositor and are never repainted by scroll (fixed layer). Phone halves their opacity and size (Appendix A) — ambience, not wallpaper.
- Route-transition animation animates `opacity` only (compositor-safe, §3.6).
- Lighthouse CI (existing `scripts/lighthouse-check.mjs`) budgets unchanged: perf + a11y on `/` mobile. If the wash/glass drops perf below budget, the fallback ladder is: remove card translucency on phone (solid surfaces) → remove orbs on phone → never remove the wash (it is one gradient). Each step logged if taken.
- **Fonts:** Inter/JetBrains/Newsreader load `display: optional` (body text — a missed swap costs nothing structural because next/font's metric-adjusted fallbacks hold the line boxes). **Playfair loads `display: "swap"`** — it sets the headlines, and `optional` would ship Georgia headlines for a whole session on a slow first load; a metric-matched fallback cannot fully mask a display serif at 46–56px. The one-frame serif swap is accepted and noted. Headline containers reserve line-height so the pre-swap fallback cannot overflow the 375px login panel; one VRT shot runs **with fonts blocked** (fallback layout locked, §9.2) — the font-loaded-only oracle was a blind spot the adversarial pass caught.

Phases R0–R6 (playbooks + gates)

Same contract as DEVELOPMENT-PLAN.md §6: TDD for logic, visual work verified by the VRT + checklist, every phase exits through the gate, tag `redesign-N`, roll straight into the next. Estimates assume the same executor pace as P0–P6.

The standing gate (every phase, in order):

```
1 npm run typecheck && npm run lint && npm test          # app unit
2 uv run pytest                                          # pipeline (phases that touch it)
3 npm run build && npx playwright test                  # e2e + VRT + PWA
4 npx lhci autorun --mobile, / --perf+ally budgets (R2+)
5 Anti-drift checklist v2 (§3.10) against fresh screenshots
  (each eyeball item applies once its surface exists -- R1 gates on the styleguide only)
6 Update PROGRESS.md · append DECISIONS/LESSONS · git tag redesign-N · push
```

R0 · Content honesty fixes [0.5–1 day]

Steps: §6.1 then §6.2, tests first throughout; `prisma migrate dev --name add_index_levels`; `prisma/seed.mjs` index values; record new FRED fixtures; extend the calendar fixture with CPI/JOBES inclusion rows; run a full fixture pipeline (`MSM_FIXTURES=1 uv run python -m jobs.job_a`) and eyeball the seeded Desk. **Exit gate additions:** label-source coupling test exists and passes; allowlist inclusion+exclusion tests pass; seeded Desk shows an S&P level $\approx$ the seeded index value with the FRED provenance line; calendar shows zero noise names from the fixture; QUESTIONS entry written if any FRED release name failed live verification.

R1 · Foundation: tokens, fonts, styleguide, guards [1.5–2 days]

Steps: (1) `globals.css` v2 from Appendix A — including the full `[data-theme="system"]` block, the `.surface*` class bodies, the `@supports` fallback, and the multi-scope `prefers-reduced-transparency` override. (2) **The atomic rename sweep, same commit:** Tailwind emits nothing for a utility whose token vanished — no build error — so the token swap and the class rename must land together. Mapping table (grep each, replace all): `bg-desk-bg → bg-paper` · `bg-academy-bg → bg-academy-paper` · `text-desk-bg → text-paper` (if present) · `rounded-edge →` per element: cards `rounded-card`, panels `rounded-panel`, controls/inputs/buttons `rounded-control`, tags/chips `rounded-chip` (the sweep is per-occurrence judgment, ~19 files; when unsure, match the §3.4/§3.5 role of the element) · `font-stretch-[...]` → delete (dead once Archivo leaves). Then run §3.10 `grep #9` — it must be empty. (3) `lib/fonts.ts` four loaders (Playfair `display:"swap"`, §7.4); bump font budget script to 560KB + per-file size printout. (4) `lib/tokens.ts` new OS constants. (5) Add `lucide-react`. (6) Build `Surface`, `AppWash` (mounting rule §4.1), restyle `SectionMasthead` (+ provenance reveal) / `StatFigure` / `Tag`, buttons/inputs recipes, `data-p2` attribute on every P2 component root. (7) Rebuild `/styleguide` to the new system incl. the contrast table + unit test and the amber-region swatch row. (8) Rewrite the anti-drift grep script to v2 (§3.10, all 11 greps) + the jsdom `data-p2` ancestor-motion test — the new greps and the new tokens land as one atomic change and must PASS on the R1 tree. (9) Stand up the VRT harness (Part 9) and capture first baselines (styleguide + login only at this phase, incl. the fonts-blocked shot). **Gate additions:** styleguide contrast unit test green; grep v2 empty (incl. #9 dead classes); fonts load offline (build precache includes all four; line-item size table printed); VRT baselines committed. **Mixed-vintage note:** between R1 and R3 the app runs new tokens under old layouts. The atomic rename keeps every existing surface *correctly colored* (both rooms included) through that window; what it looks like is the old geometry in the new palette — acceptable, tracked, and burned down room by room in R2–R5.

R2 · Shell: nav, tab bar, wash, login, PWA chrome, palette [1–1.5 days]

Steps: AppWash into both room layouts (inside the theme scope, §4.1) + `relative z-10` wrappers; top bar restyle + newly-sticky verification (§4.2); D2 bottom tab bar + z-ladder + keyboard hide + safe areas (top/bottom/left/right) + content padding; route-transition `template.tsx` (opacity-only, §3.6) in both rooms; `CommandPalette` restyle + typed result groups + ⌘K hint chip + phone search trigger + visualViewport cap; login split-screen (§5.7); theme-color-from-cookie + Academy viewport export + manifest/status-bar/icon set (§7.3); OfflineRibbon restyle. **Gate additions:** `nav.spec.ts` extended (tab bar cases per D2 incl. keyboard-hide) green on phone project; `auth.spec.ts` / `pwa.spec.ts` / `offline.spec.ts` green (login stayed static, icons resolve, SW behavior unchanged); theme-color e2e (§7.3); safe-area checks per §7.2; the iOS manual checklist run and photographed; VRT: login + shell chrome both themes.

R3 · The Desk [1.5–2 days]

Steps: Desk header; broadsheet-spread grid with ritual DOM order + compact CalendarRow variant; module-by-module restyle per §5.1 (Pulse w/ R0 content, Brief + held-state skeleton, Calendar + quiet-card empty state, Movers + quiet line, Watchlist, SetupCards — collapsed summary stripped to `n=` + tier per P4, proportion bar moved inside `BaseRate`, instant `<details>` expansion — Scorecard/Journal); RailDialog + sheet + grab handle; sparkline upgrade; placeholders + shimmer. **Gate additions:** `desk.spec.ts` / `drill.spec.ts` / `briefing.spec.ts` / `setup-cards.spec.ts` green with updated styling-level assertions; §3.10 `grep #6 + #10` + the `data-p2` ancestor test green (the `<details>`-instant rule is covered by the ancestor test); VRT: Desk seeded, rail open, both themes, both viewports.

R4 · Sub-rooms: Scans, Paper, Track record, Ticker + Range Ladder [1.5–2 days]

Steps per §5.2–§5.5. `RangeBands` geometry is TDD'd as a pure function (band + dotplot coordinates from vol-band data); the SVG renders the computed geometry; the receipt (§5.3) and recipe-card scans (§5.2) land here. `QuantileDotplot` ships here (the P6-deferred widget, now in-system). **Gate additions:** `paper.spec.ts` + track-record assertions green; Range Ladder locks: a DOM-level unit assertion that no path/line/tick exists at the 50th percentile and no element connects band rows (the no-median, no-cone locks); VRT: all four rooms, both themes/viewports.

R5 · The Academy [1–1.5 days]

Steps per §5.6 (solid warm cards, serif kickers, module palette, reading typography, review ritual, glossary-as-hero, quote card; phone top-bar doorway drop per §4.2). Academy stays light-only; `theme.spec.ts` assertion kept + Academy theme-color case. **Gate additions:** `academy.spec.ts` green; no-price-in-Academy check unchanged; VRT: academy home/lesson/review/glossary, phone + desktop.

R6 · Hardening & drift-proofing [1 day]

Steps: touch-target sweep (§7.1); axe pass on every route (existing e2e axe hooks); Lighthouse budgets; leftover-surface grep (§4.1) + full §3.10 including dead-class re-check; iOS manual checklist re-run; full VRT re-baseline review; kill dead tokens/classes; mint skills (`new-surface`, `vrt-update` — Part 10); final docs sync (regenerate DEVELOPMENT-PLAN.md + PDFs if any §3 text drifted during build); PROGRESS/DECISIONS/LESSONS closeout; tag `redesign-final`. **Gate:** the full standing gate + every §3.10 eyeball item photographed into `docs/redesign-evidence/` (the same screenshot-evidence habit as P6).

Sequencing rules: R0 blocks everything (honesty first). R1 blocks R2–R5. R2 blocks R3 (shell before rooms). R3–R5 are order-fixed (Desk → sub-rooms → Academy) so shared primitives harden in the highest-traffic room first. A D-decision veto changes only its named sections; none blocks R0/R1 except D1 (dark column) and D4 (academy column) which land in R1's token sheet — if either veto arrives after R1, it is a token-column edit, not a rework.

Visual regression (new instrument; the styling counterpart of TDD)

Today no VRT exists (the styleguide comments reference an aspirational suite). The redesign is exactly when it becomes real: styling this large without pixel locks would drift within a week.

9.1 Harness

Playwright `toHaveScreenshot` inside the existing e2e setup (same prod-build webServer, same two projects: `desktop` 1366×768 Chromium, `phone` Pixel 7). New spec file `e2e/vrt.spec.ts` + config additions:

- `expect.toHaveScreenshot` defaults: `maxDiffPixelRatio: 0.01`, `animations: "disabled"`, `caret: "hide"`, `scale: "css"`.
- Every VRT page load forces `prefers-reduced-motion: reduce` (kills route transitions and entrance fades → deterministic pixels) and waits for the fonts via `await page.evaluate(async () => { await document.fonts.ready; })` (returning the bare `FontFaceSet` from `evaluate` serializes to junk — the `await` must happen inside the page; Appendix D shows the exact pattern). One designated shot per §9.2 runs **with fonts blocked** to lock the fallback-metrics layout (§7.4).
- **Masking**: timestamps and as-of stamps get `data-vrt="mask"`; the spec collects `[data-vrt=mask]` into the `mask` option. Seeded fixture data (`MSM_SEEDED=1`) makes numbers stable; stamps are masked because they encode wall-clock time.
- Baselines live in `e2e/__screenshots__` and are committed. CI runs Linux — pixel fonts differ from macOS, so baselines are generated IN CI (the standard Playwright `--update-snapshots` bootstrap on the first CI run of a phase) or via the docker runner; local macOS runs use `--ignore-snapshots` unless the developer regenerates a local set. Decision logged (Appendix E-7); simplest workable policy first: CI is the pixel oracle.

9.2 The shot list

Page/state	Projects	Themes
/styleguide — per-section (<code>tokens</code> , <code>surfaces</code> , <code>type</code> , <code>chips</code> , <code>controls</code> , <code>charts</code> , <code>motion-static</code>)	both	light + dark
/login	both	light
/ seeded (header + all modules)	both	light + dark
/ rail open (watchlist drill)	both	light
/scans seeded	both	light + dark
/paper (form + receipt + interstitial open)	both	light + dark
/track-record seeded	both	light + dark
/ticker/[seed symbol] (candle + FanChart)	both	light + dark
/academy home · lesson · review · glossary	both	light (Academy has no dark)
bottom tab bar states (per D2)	phone	light + dark
rail bottom sheet open (grab handle + inset)	phone	light
/login with fonts BLOCKED (fallback layout lock, §7.4)	phone	light
/ landscape (tab bar + side insets — browser tabs rotate)	phone (landscape viewport)	light

(~50 baselines; each phase captures only its own rows — the table completes at R5.)

9.3 Update protocol (the drift-vs-intent discipline)

An intentional restyle updates baselines in the same PR with a one-line note in the commit body ("VRT: N baselines updated — <reason>"). An UNEXPLAINED diff is a build failure, full stop. Phase gates re-run the whole accumulated table, not just the phase's rows — R5 proves the Desk still matches R3's pixels. The eyeball step in the gate (§3.10 checklist) reviews the diff images, not just the pass/fail.

Docs, decisions, and skills (executed alongside this plan)

The constitution edits this plan mandates are performed in the same session that authored it, so no successor session ever reads two constitutions. The list is the receipt. This plan was revised after a four-lens adversarial review — executor-stall (16 findings, every claim checked against the working tree), honesty-leakage (20), mobile/iOS (12), and traced-vs-designed (11) — and the fixes are integrated inline rather than appended; where a finding reversed an earlier call, the section says so.

-
1. **CLAUDE.md**: design one-liner replaced with §1.4; the non-negotiables digest line reworded to "general UI motion allowed; probability/money visuals never move"; commands and rituals untouched.
 2. **DEVELOPMENT-PLAN.md Part 3**: edited in its source (`docs/src/dp-0*.html`) — §3.1 banned-list and manifesto, §3.3 palette pointer, §3.4 shape/elevation, §3.5 motion, §3.10 greps — each amended clause gets an `Amended 2026-07-11` marginal note pointing here; the markdown regenerated via `docs/src/build-plan-md.py`; `docs/Development-Plan.pdf` re-rendered. Appendix-K CLAUDE.md template mirror updated to match (plan lines ~924–956).
 3. **Research Report §9.7/§9.8** (`docs/src/rr-04.html` + the combined `research-report.html`): aesthetic clauses amended with a clearly-attributed dated amendment box (user directive 2026-07-11); honesty clauses untouched; `docs/Research-Report.pdf` re-rendered. The report's evidence chapters are not touched — only the §9.7/9.8 design-language clauses.
 4. **DECISIONS.md**: structural entry superseding the line-20 "token-for-token Broadsheet Terminal" decision; entries for the local decisions in Appendix E.
 5. **QUESTIONS-FOR-BISHANT.md**: the four D-decisions mirrored as `[NEED]` items (they are also the top of this plan; the file is the durable copy).
 6. **This document**: `UI-REDESIGN-PLAN.md` at the repo root; typeset copy at `docs/UI-Redesign-Plan.pdf` from `docs/src/ui-redesign-plan.html` (print.css pipeline).
 7. **Skills to mint during the build**: `vrt-update` (the §9.3 protocol as a checklist), `new-surface` (Surface/masthead/chip recipes for any future module). Mint at first use (R1/R3), per the existing skill policy.

`globals.css` v2 (paste-ready)

The full replacement token sheet + the surface/wash class bodies. Comments are part of the deliverable (the sheet is read by humans first — house rule). Values match Part 3; if any pair fails the R1 contrast test, darken the foreground token one step and log it. NOTE: this sheet is paste-COMPLETE — the dark and system blocks are both written out in full below (they must stay in lockstep; a stub here once shipped an empty system block in review).

```

/*
 * globals.css – the single source of every design token in myStockMarket.
 *
 * The design system is "Morning Broadsheet" (UI-REDESIGN-PLAN.md Part 3): editorial serif
 * over mono numerals, a lavender morning-light wash, glass cards with soft depth, hairlines
 * inside cards, one hero figure per view, color scarce and always meaningful.
 *
 * HARD RULE (plan $3.10 v2, grepped at every phase exit): no hex color, raw gradient,
 * arbitrary shadow, or arbitrary radius may appear anywhere else in the codebase
 * (lib/tokens.ts holds the three OS-facing constants). If you need a value, it is already
 * here, or it does not belong in the product.
 *
 * Tailwind v4 caveat: rgb(r g b / a) tokens work as utilities (bg-surface) but the
 * slash-opacity modifier (bg-surface/50) silently no-ops on them. Never use it; mint a token.
 */

@import "tailwindcss";

@theme static {
  /* — Type families (§3.1) – four, no more ————— */
  --font-display: var(--font-playfair), ui-serif, Georgia, serif; /* titles; never numbers,
                                                                    never body, never <19px */
  --font-ui: var(--font-inter), ui-sans-serif, system-ui, sans-serif;
  --font-mono: var(--font-jetbrains), ui-monospace, "SF Mono", Menlo, monospace;
  --font-prose: var(--font-newsreader), ui-serif, Georgia, serif; /* prose + small italics */

  /* — Type scale (§3.2) – existing pairs kept, four additions ————— */
  --text-2xs: 11px; --text-2xs--line-height: 1.35;
  --text-xs: 12px; --text-xs--line-height: 1.4;
  --text-sm: 13.5px; --text-sm--line-height: 1.5;
  --text-base: 15px; --text-base--line-height: 1.55;
  --text-input-touch: 16px; /* every form control below md (iOS zoom) */
  --text-prose: 17.5px; --text-prose--line-height: 1.65;
  --text-title: 19px; --text-title--line-height: 1.3; /* serif card titles (floor) */
  --text-lg: 21px; --text-lg--line-height: 1.35;
  --text-xl: 27px; --text-xl--line-height: 1.25;
  --text-2xl: 34px; --text-2xl--line-height: 1.15;
  --text-display: clamp(30px, 4vw, 46px); --text-display--line-height: 1.1;
  --text-display-hero: clamp(30px, 5vw, 56px); --text-display-hero--line-height: 1.05;
  --text-num-lg: 44px; --text-num-lg--line-height: 1.1;
  --text-hero: 64px; --text-hero--line-height: 1;
  --text-hero-mobile: 48px; --text-hero-mobile--line-height: 1;

  /* — Color: Desk, light ("Morning") (§3.3) ————— */
  --color-paper: #f9f8ff; /* page plane under the wash */
  --color-surface: rgb(255 255 255 / 0.72); /* L1 card (no blur on phones) */
  --color-surface-raised: rgb(255 255 255 / 0.90); /* L2 */
  --color-surface-bar: rgb(249 248 255 / 0.85); /* L3 sticky bars (with blur) */
  --color-surface-overlay: rgb(255 255 255 / 0.96); /* L4 overlays (with blur) */
  --color-surface-solid: #ffffff; /* fallbacks + decision moments */
  --color-ink: #0f0d1a;
  --color-ink-2: #3d3a4f;
  --color-muted: #6e6c80; /* provenance – must clear 4.5:1 on paper */
  --color-faint: #a9a7b8; /* placeholders/disabled; never body text */
  --color-hairline: rgb(99 98 143 / 0.13);
  --color-hairline-strong: rgb(99 98 143 / 0.24);

  /* Accent – interactive only; never on data, never on status (§3.3) */
  --color-accent: #6366f1;
  --color-accent-deep: #4f46e5; /* link/label text on light */
  --color-accent-soft: rgb(99 102 241 / 0.10);
  --color-accent-muted: rgb(99 102 241 / 0.06);

  /* Band – non-directional uncertainty fills ONLY (range ladder). Grayer than accent
   * on purpose: an uncertainty band must never look clickable or directional. */
  --color-band: #6d648c;
  --color-band-outer: rgb(109 100 140 / 0.10);
  --color-band-inner: rgb(109 100 140 / 0.22);
  --color-band-panel: rgb(109 100 140 / 0.14); /* the flat typical-range fill */

  /* Semantic pair – data only, always redundantly encoded (§3.3) */
  --color-up: #2563eb; --color-up-text: #1d4ed8; --color-up-wash: rgb(37 99 235 / 0.08);
  --color-down: #ea580c; --color-down-text: #c2410c; --color-down-wash: rgb(234 88 12 / 0.08);
  --color-up-bar: rgb(37 99 235 / 0.35); /* proportion/breadth segments – equal weight */
  --color-down-bar: rgb(234 88 12 / 0.35);

  /* Reserved amber – the REGION is reserved, not just the hex (§3.3): only losses and
   * these two consumers (gate flag, fired-signal) may sit in the amber-orange range. */
  --color-alert: #8a5200;
  --color-alert-wash: #f7ead0;

  /* Tier chips (§3.3) – word always inside the chip; moderate is TEAL, not amber */
  --color-tier-strong: #047857; --color-tier-strong-wash: rgb(5 150 105 / 0.10);
  --color-tier-moderate: #0f766e; --color-tier-moderate-wash: rgb(13 148 136 / 0.10);
  --color-tier-weak: #6e6c80; --color-tier-weak-wash: rgb(124 122 142 / 0.10);

  /* Evidence grades – inside Tag only, never without their word; mixed is teal */
  --color-grade-supported: #047857;
  --color-grade-mixed: #0f766e;

```

```

--color-grade-weak: #a13d22;
--color-grade-folklore: #b91c1c;

/* Academy (warm room, D4) – solid paper, no glass (§5.6); interactives stay indigo */
--color-academy-paper: #fbf6ee;
--color-academy-surface: #ffffd8; /* SOLID – the reading room isn't glass */
--color-academy-hairline: rgb(143 110 72 / 0.15);

/* Academy module hues – DECORATIVE ONLY (§5.6); none amber, none the accent */
--color-module-foundations: #4a5a8c;
--color-module-structure: #5b7a4f;
--color-module-patterns: #7a4a6b;
--color-module-risk: #9c4460;

/* — Gradients (§3.4) – the only four; raw gradients grepped elsewhere — */
--gradient-brand: linear-gradient(135deg, #4f46e5 0%, #7c3aed 100%);
--gradient-wash: linear-gradient(145deg, #f9f8ff 0%, #eef2ff 35%, #f0f4ff 65%, #faf8ff 100%);
--gradient-spark-up: linear-gradient(180deg, rgb(37 99 235 / 0.12) 0%, rgb(37 99 235 / 0) 100%);
--gradient-spark-down: linear-gradient(180deg, rgb(234 88 12 / 0.12) 0%, rgb(234 88 12 / 0) 100%);

/* Orbs (L0 wash depth) – re-derived for the real product (§1.3 #14): desktop values;
 * phone halves size+opacity via the media block below */
--orb-1: radial-gradient(circle, rgb(139 92 246 / 0.08) 0%, transparent 70%);
--orb-2: radial-gradient(circle, rgb(99 102 241 / 0.06) 0%, transparent 70%);

/* — Elevation (§3.4) — */
--shadow-soft: 0 4px 24px rgb(99 88 143 / 0.08), 0 1px 4px rgb(0 0 0 / 0.04);
--shadow-lift: 0 6px 24px rgb(99 88 143 / 0.12);
--shadow-overlay: 0 8px 40px rgb(99 88 143 / 0.15);

/* — Shape (§3.5) — */
--radius-card: 16px;
--radius-panel: 12px;
--radius-control: 10px;
--radius-chip: 8px;
--radius-pill: 999px;

/* — Motion (§3.6) – probability/money visuals never consume these, nor their
 * ancestors: every P2 component root carries data-p2 and a jsdom test walks up. — */
--duration-quick: 150ms;
--duration-fade: 200ms;
--duration-slide: 240ms;
--ease-quiet: cubic-bezier(0, 0, 0.2, 1);

/* — Layout — */
--breakpoint-desk: 1366px;
}

:root {
  --scrim: rgb(15 13 26 / 0.4);
}

/* — Midnight Desk (D1) – Desk-only via [data-theme]; Academy never darkens —
 * The [data-theme="system"] block below MUST stay byte-identical to this one. */
[data-theme="dark"] {
  --color-paper: #14121f;
  --color-surface: rgb(30 27 48 / 0.72);
  --color-surface-raised: rgb(36 32 56 / 0.90);
  --color-surface-bar: rgb(20 18 31 / 0.85);
  --color-surface-overlay: rgb(30 27 48 / 0.96);
  --color-surface-solid: #1e1b30;
  --color-ink: #ece9f8;
  --color-ink-2: #c3c0d6;
  --color-muted: #918ea6;
  --color-faint: #6b6880;
  --color-hairline: rgb(167 157 210 / 0.16);
  --color-hairline-strong: rgb(167 157 210 / 0.28);
  --color-accent: #818cf8;
  --color-accent-deep: #a5b4fc;
  --color-accent-soft: rgb(129 140 248 / 0.14);
  --color-accent-muted: rgb(129 140 248 / 0.08);
  --color-band: #8f86b8;
  --color-band-outer: rgb(143 134 184 / 0.14);
  --color-band-inner: rgb(143 134 184 / 0.28);
  --color-band-panel: rgb(143 134 184 / 0.18);
  --color-up: #60a5fa; --color-up-text: #93c5fd; --color-up-wash: rgb(96 165 250 / 0.12);
  --color-down: #fb923c; --color-down-text: #fdb474; --color-down-wash: rgb(251 146 60 / 0.12);
  --color-up-bar: rgb(96 165 250 / 0.38);
  --color-down-bar: rgb(251 146 60 / 0.38);
  --color-alert: #e0a83e; --color-alert-wash: #3a2f14;
  --color-tier-strong: #4cbf7e; --color-tier-strong-wash: rgb(76 191 126 / 0.14);
  --color-tier-moderate: #45c7b8; --color-tier-moderate-wash: rgb(69 199 184 / 0.14);
  --color-tier-weak: #918ea6; --color-tier-weak-wash: rgb(145 142 166 / 0.14);
  --color-grade-supported: #4cbf7e;
  --color-grade-mixed: #45c7b8;
  --color-grade-weak: #d07b5b;
  --color-grade-folklore: #e06c6c;
  --gradient-wash: linear-gradient(145deg, #14121f 0%, #191627 40%, #151329 70%, #14121f 100%);
  --gradient-spark-up: linear-gradient(180deg, rgb(96 165 250 / 0.14) 0%, rgb(96 165 250 / 0) 100%);
  --gradient-spark-down: linear-gradient(180deg, rgb(251 146 60 / 0.14) 0%, rgb(251 146 60 / 0) 100%);

```

```

--orb-1: radial-gradient(circle, rgb(139 92 246 / 0.05) 0%, transparent 70%);
--orb-2: radial-gradient(circle, rgb(99 102 241 / 0.04) 0%, transparent 70%);
--shadow-soft: 0 4px 24px rgb(0 0 0 / 0.45), 0 1px 4px rgb(0 0 0 / 0.3);
--shadow-lift: 0 6px 24px rgb(0 0 0 / 0.5);
--shadow-overlay: 0 8px 40px rgb(0 0 0 / 0.5);
}
@media (prefers-color-scheme: dark) {
  [data-theme="system"] {
    /* FULL duplicate of the [data-theme="dark"] block above – the executor writes it out
       verbatim (all ~45 lines). Keep the two in lockstep; a drift test in R1 compares them
       by parsing this file. */
  }
}

/* — Behavior globals (kept from v1, values re-pointed) ————— */
html {
  background-color: var(--color-paper);
  color: var(--color-ink);
  font-variant-numeric: tabular-nums lining-nums;
  -webkit-text-size-adjust: 100%;
  overflow-x: hidden; /* the page never scrolls sideways */
  overscroll-behavior-y: contain; /* no pull-to-refresh surprise in standalone (§7.1) */
}
body { font-family: var(--font-ui); font-size: var(--text-sm); line-height: var(--text-sm--line-height); }

/* L0 wash – fixed layer, never background-attachment (iOS). MUST be mounted INSIDE the
 * data-theme / academy scope or it reads the wrong wash (§4.1). */
.app-wash { position: fixed; inset: 0; z-index: 0; pointer-events: none;
  background: var(--gradient-wash); transform: translateZ(0); }
.app-wash::before, .app-wash::after { content: ""; position: absolute; border-radius: 999px; }
.app-wash::before { top: -140px; right: -100px; width: 620px; height: 620px; background: var(--orb-1); }
.app-wash::after { bottom: -120px; left: -80px; width: 420px; height: 420px; background: var(--orb-2); }
@media (max-width: 767px) {
  .app-wash::before { width: 340px; height: 340px; opacity: 0.6; }
  .app-wash::after { width: 240px; height: 240px; opacity: 0.6; }
}

/* Surfaces (§3.4) – L1/L2 never blur; L3/L4 blur on an inner wrapper so iOS clips the
 * blur to the radius (Safari does not clip backdrop-filter to border-radius). */
.surface { background: var(--color-surface); border: 1px solid var(--color-hairline);
  border-radius: var(--radius-card); }
.surface-raised { background: var(--color-surface-raised); border: 1px solid var(--color-hairline);
  border-radius: var(--radius-card); box-shadow: var(--shadow-soft); }
.surface-bar { background: var(--color-surface-bar); }
.surface-bar > .bar-blur { backdrop-filter: blur(20px); -webkit-backdrop-filter: blur(20px);
  overflow: hidden; }
.surface-overlay { background: var(--color-surface-overlay); border: 1px solid var(--color-hairline-strong);
  border-radius: var(--radius-panel); box-shadow: var(--shadow-overlay); overflow: hidden;
  backdrop-filter: blur(24px); -webkit-backdrop-filter: blur(24px); }
@supports not (backdrop-filter: blur(1px)) {
  .surface-bar, .surface-overlay { background: var(--color-surface-solid); }
}

.no-scrollbar { scrollbar-width: none; -webkit-overflow-scrolling: touch; }
.no-scrollbar::-webkit-scrollbar { display: none; }

:focus-visible { outline: 2px solid var(--color-accent); outline-offset: 2px; }

@media (prefers-reduced-motion: reduce) {
  *, ::before, ::after {
    animation-duration: 0ms !important; animation-iteration-count: 1 !important;
    transition-duration: 0ms !important; scroll-behavior: auto !important;
  }
}

/* Reduced transparency must win at EVERY theme scope – a :root-only override loses to the
 * theme blocks above (custom properties resolve at the nearest defining ancestor). */
@media (prefers-reduced-transparency: reduce) {
  :root, [data-theme="dark"], [data-theme="system"] {
    --color-surface: var(--color-surface-solid);
    --color-surface-raised: var(--color-surface-solid);
    --color-surface-bar: var(--color-surface-solid);
    --color-surface-overlay: var(--color-surface-solid);
  }
}

```

(The Academy layout applies its room by re-declaring `--color-paper: var(--color-academy-paper)`, `--color-surface: var(--color-academy-surface)`, `--color-hairline: var(--color-academy-hairline)`, and its orb/wash tokens on the academy shell element — the same scoped-override mechanism as dark mode; the reduced-transparency block above already covers it because the academy surface is solid by design.)

copy.ts additions (mechanical voice; exact strings)

Key	String
macro.provenance	Index levels · FRED · prior close
macro.proxyChip	ETF proxy
macro.proxyNote	{symbol} is an ETF whose price tracks this group; it is not the index level.
macro.indexUnavailable	Index level unavailable – showing {symbol} (ETF proxy).
desk.edition	The Desk – Evening Edition
desk.status	Markets {state} · data as of {close} · updated {stamp}
calendar.importanceHigh	high
calendar.empty	A quiet stretch – no scheduled catalysts in the next 14 days.
calendar.emptySub	Curated from the full FRED release feed; the rest was noise.
mover.quiet	No moves cleared the catalyst-or-noise bar today.
mover.revolNote	RelVol = volume ÷ 20-day average. "No clear catalyst" usually means noise; a cause sometimes surfaces later – absence of news is not a reason
palette.hint	⌘K
palette.placeholder	Go to a room, lesson, or ticker...
login.headline	Your personal broadsheet for the market.
login.subline	Not a prediction oracle. Not a signal feed. A daily record of what happened, why it might matter, and what the base rates actually say – inclu
login.quote	"An investor who has all the answers doesn't even understand the questions."
login.quoteAttribution	– Sir John Templeton
academy.quote	"The first step to understanding markets is understanding that no one fully understands markets."
dotplot.caption	Each dot: 1 in 20 past outcomes.
dotarray.caption	Each dot is one historical case. Misses are intentionally visible.

The Range Ladder reuses the pinned `volband.label` + `volband.caveat` verbatim (per-horizon frequency sentence + regime caveat) — no parallel fan-label strings exist, so the chart and its accessible table can never drift apart. `calendar.noEdge` is untouched (it means "no directional edge", not "no events"). Rules unchanged: components never inline strings; `fill()` still throws on placeholder mismatch; the copy unit test pins every string above.

FRED release allowlist (Bug 2) and index series (Bug 1)

Series (Bug 1): `SP500` (S&P 500, daily close, ~1-day lag), `NASDAQCOM` (Nasdaq Composite), `DJIA` (Dow Jones Industrial Average) — level + prior via `latest_two`. Russell 2000: no free FRED series — IWM stays an honest proxy (Appendix E-1). UI labels: "S&P 500", "Nasdaq Composite" (NOT "Nasdaq 100"), "Dow". Fallback proxy labels per slot: `SPY`, `QQQ` labeled "Nasdaq-100" (it does not track the Composite), `DIA`.

Release allowlist (Bug 2) — the single source of the calendar vocabulary; `code` is what the chip renders:

FRED release name (match)	code	display	kind	importance
Consumer Price Index	CPI	Consumer Price Index	macro	high
Employment Situation	JOBS	Jobs report	macro	high
Producer Price Index	PPI	Producer Price Index	macro	medium
Gross Domestic Product	GDP	GDP	macro	medium
Personal Income and Outlays	PCE	PCE (inflation)	macro	medium
Advance Monthly Sales for Retail and Food Services	RETAIL	Retail sales	macro	medium
FOMC Press Release	FOMC	FOMC decision	fed	high

FMP earnings rows: `code: EARNINGS`, importance high when the symbol is in `_CORE_SERVED`, else medium. Explicitly excluded despite sounding relevant: "Standard & Poors" and "Dow Jones Averages" (index-data publications, not catalysts), "H.15 Selected Interest Rates" (daily), "CBOE Market Statistics" (daily). Everything not on the list is dropped — the list only ever grows by a deliberate commit.

VRT config sketch (Part 9 made concrete)

```
// playwright.config.ts – additions (top-level expect block; projects already match $9.1)
expect: {
  toHaveScreenshot: {
    maxDiffPixelRatio: 0.01,
    animations: "disabled",
    caret: "hide",
    scale: "css",
  },
},

// e2e/vrt.spec.ts – the pattern every shot uses
test.use({ contextOptions: { reducedMotion: "reduce" } });
async function shoot(page: Page, path: string, name: string) {
  await page.goto(path);
  await page.evaluate(async () => { await document.fonts.ready; }); // await INSIDE the page
  const masks = page.locator('[data-vrt="mask"]');
  await expect(page).toHaveScreenshot(`${name}.png`, { fullPage: true, mask: [masks] });
}
// fonts-blocked shot: context.route('**/*.woff2', r => r.abort()) before goto, skip fonts.ready
```

Dark-theme shots set the theme cookie before navigation (same mechanism as `theme.spec.ts`). Seeded shots gate on `MSM_SEEDED=1` like the existing data specs.

Logged local decisions (the designer's log; each also lands in DECISIONS.md)

1. **Russell 2000 hybrid**: index levels for SPX/Nasdaq Composite/Dow from FRED; small-caps slot keeps IWM as an explicit, chip-labeled ETF proxy. Rejected: dropping small caps (loses breadth-of-market at a glance); a new data provider (out of scope).
2. **Fonts**: Playfair Display (display) + Inter (UI) + JetBrains Mono (numerals) + Newsreader (prose + small editorial italics, kept). Serif floor: Playfair $\geq 19\text{px}$, never italic below display sizes. Rejected: Playfair for prose or small italics (display face, fragile hairlines at text sizes); dropping Newsreader (the Academy is a reading room).
3. **Gain/loss pair** moves Wong $\rightarrow$ Figma blue/orange (#2563eb/#ea580c + AA text steps). Both are colorblind-safe; the Figma pair sits naturally in the indigo world. Redundant encoding unchanged.
4. **Earnings importance heuristic**: high when the symbol is in the pipeline's served core, else medium. The user-watchlist half of the first draft was dropped in review — the watchlist is app-DB state the ingest doesn't hold. Rejected: consensus-size heuristics (false precision).
5. **Desk two-column at 1366px as a broadsheet spread** — narrative modules in sequence in the main column, reference modules (Watchlist, compact Calendar, SourceStatus) in the rail; DOM and phone keep the strict ritual order. Rejected: reordering DOM (breaks tab order and the phone ritual); single-column desktop (wastes the width the one-screen claim was written for).
6. **<details> stays** for setup cards (semantics + zero JS), with the expansion instant because its subtree is P2. Rejected: extending the symbol-fixed Rail (P6 decision), animated expansion (motion over probability content).
7. **VRT pixel oracle is CI (Linux)**; local macOS runs ignore snapshots unless regenerated. Rejected: per-OS baseline sets (doubles maintenance for one developer).
8. **Route transitions via CSS `template.tsx`, opacity-only** — no translateY (review: a translating page moves the probability visuals inside it; a uniform fade contains no relative motion), not the experimental View Transitions flag (unstable API — a stall risk).
9. **Theme toggle stays on /settings**, labeled "Desk theme" with its scope in helper text. Rejected: nav toggle (chrome creep), unscoped "Dark mode" label (lies about the Academy).
10. **Login brand-panel copy** enters `copy.ts` (login.* keys) — even marketing-flavored lines obey the one-deck rule and the mechanical voice (attributed quote allowed).
11. **Mastheads stay muted, accent only on interactive text + small in-card slot labels** — matching what the Figma actually does (its eyebrows are #7c7a8e, its accents are the small slot headings). The first draft accented the mastheads and was reversed in review: eight indigo mastheads would make the accent mean "chrome" as often as "act here".
12. **Figma's "Focus Watchlist" 14-day sparkline window** adopted — historical prices in a sparkline are a record, not a forecast; allowed and wanted.
13. **The amber–orange REGION is reserved, not just the alert hex** — tier-moderate and grade-mixed moved to teal; module hues chosen off-amber. Reservation is perceptual: a Desk of amber-ish chips would drown the gate flag with zero hex collisions.
14. **The Range Ladder replaces the fan-cone** — per-horizon rows on a return axis, nested 50/80 bands, embedded quantile dotplots, no median mark, no connecting silhouette. A forward-widening cone is the visual grammar of a projection; the ladder is the honest fan, sliced. Bands paint in `--color-band` (new), never the accent (accent = interactive) and never up/down (a symmetric range has no direction).
15. **Atomic class rename in R1** (option b) instead of legacy token aliases — Tailwind v4 silently drops utilities whose tokens vanish, so aliases would hide drift; the rename + grep #9 make the cut visible and complete in one commit. Rejected: aliases until R6 (invisible rot).
16. **The Academy diverges structurally, not just chromatically** — solid paper cards (no glass), serif kickers instead of terminal mastheads, reading line-height. The first draft's "same glass, warmer hexes" was rightly called a tinted dashboard in review.
17. **Charts appear as completed wholes** (200ms fade max); no directional draw-in anywhere — a left-to-right sweep on a price series reads as momentum next to visuals that insist they are not forecasts.

UI Redesign Plan compiled July 2026 for executor Claude Opus 4.8; revised after a four-lens adversarial review (executor-stall, honesty-leakage, mobile/iOS, traced-vs-designed). The markdown working copy at `UI-REDESIGN-PLAN.md` is the source of truth; this typeset copy is generated from it.